

A study on state estimation of distributed electric drive articulated vehicle

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Abstract

In view of the high direct measurement cost and high complexity of dynamic model in state estimation of distributed electric drive articulated vehicles, this study proposes a more efficient and accurate sideslip angle and yaw rate estimation method suitable for distributed electric drive articulated vehicles. First, four degrees of freedom and six degrees of freedom dynamic models are constructed, and two state estimators are designed based on extended Kalman filter (EKF) and unscented Kalman filter (UKF), respectively. The estimation performance of different models and algorithms under sine and fishhook conditions is compared and analyzed. The study shows that under low-speed weak nonlinear conditions, the four degrees of freedom model combined with EKF can improve computational efficiency while ensuring accuracy, while under high-speed strong nonlinear conditions, the six degrees of freedom model combined with UKF can achieve the best estimation effect. In addition, the feasibility of the UKF state estimator based on the 6DOF model is verified by the Logitech G27 driving simulator. The study lays the foundation for subsequent multi-model fusion and real-time optimization.

Keywords: Dynamic, state estimation, articulated vehicles

Nomenclature

u_1/u_2 :	Longitudinal velocity of the unit 1/unit 2 (km/h)
v_1/v_2 :	Lateral velocity of the unit 1/unit 2 (km/h)
β_1/β_2 :	Sideslip angle of the unit 1/unit 2 (deg)
ω_1/ω_2 :	Yaw rate of the unit 1/unit 2 (deg/s)
R_1/R_2 :	Distance from the CG of the unit 1/unit 2 to the instantaneous turning center (m)
δ_i :	Steering angle of the i axle (deg)
ϕ :	Angle at hinge joint (deg)
a_{y1} :	Lateral acceleration of the unit 1(m/s^2)

1. Introduction

With the rapid development of distributed electric drive technology, distributed electric drive articulated vehicles have been widely used in specific industrial scenarios such as tunnels, harbors, and mines. At the same time, with the continuous breakthroughs in unmanned driving technology, distributed electric drive articulated vehicles are also developing towards intelligence, which has put forward an urgent need for high-precision state perception (Korayem et al., 2021). The characteristics of distributed electric drive articulated vehicles such as long body, high center of gravity and the presence of articulated devices make the vehicle dynamics model more complex and have stronger nonlinear characteristics. When calculating, not only the dynamic influence of the front and rear bodies must be considered, but also the coupling effect of the change of the articulation angle. However, ac-

curate estimation of the sideslip angle and yaw rate is crucial to improving vehicle stability control, optimizing steering strategies and improving the reliability of intelligent driving systems. It can effectively reduce the risk of rollover and improve the handling performance of the vehicle. Therefore, it is of great significance to carry out research on the estimation of the sideslip angle and yaw rate for distributed electric drive articulated vehicles (Liu et al., 2023; Wu et al., 2021).

Vehicle state parameter estimation primarily follows two approaches: model-based estimation and data-driven estimation. To enhance the accuracy of sideslip angle estimation for three-axle vehicles, Chen et al. (2024) introduced a fusion estimation technique employing an adaptive cubature Kalman filter, demonstrating significant improvements in estimation precision. Meanwhile, Xia et al. (2022) developed a multi-model information fusion strategy to estimate the joint angle of semi-trailer trains across diverse driving conditions, further optimizing the algorithm through genetic algorithms. For all-wheel-drive electric vehicles, Heidfeld & Schunemann (2021) leveraged optimization-based tuning techniques and unscented Kalman filtering to achieve highly precise state parameter estimation. Addressing four-wheel independent drive electric vehicles, Ding et al. (2020) presented a longitudinal speed estimator based on multi-sensor fusion, which was rigorously validated through hardware-in-the-loop testing to ensure accuracy and reliability. In an effort to refine rolling resistance estimation, Sharma et al. (2021) proposed an unknown input adaptive high-gain observer, experimentally verified to maintain an average relative error below 10%. Safety-critical vehicle state es-

Received: March 28, 2025. Revised: June 26, 2025. Accepted: June 27, 2025

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timation has also been extensively explored. For instance, Liu *et al.* (2021) designed a safety attitude estimator employing the unscented Kalman filter (UKF), with simulation results validating its effectiveness. Similarly, Li *et al.* (2020) constructed a vehicle state estimation framework incorporating Cubature Kalman Filter (CKF)-based estimators for driving state, road adhesion coefficients, and vehicle parameters, ultimately forming a closed-loop system validated via a driving simulator. Expanding on sideslip angle estimation techniques, Bascetta *et al.* (2020) introduced two novel methods rooted in kinematic modeling, both tested under various operational scenarios. For distributed electric vehicles, Jin *et al.* (2023) developed a dual-center differential Kalman filter, enabling simultaneous estimation of inertial parameters and dynamic characteristics via two parallel Kalman filters. A hybrid Kalman filtering approach was adopted by Rodríguez *et al.* (2021), integrating extended Kalman filter (EKF) and UKF to refine vehicle dynamics estimation. Moreover, Xia *et al.* (2023) focused on balanced forklifts, proposing a Center of Gravity (CG) estimation technique that was verified on a hardware-in-the-loop (HIL) platform. To enhance adaptability in sensor failure scenarios, Yuan & Song (2023) designed an adaptive correction EKF algorithm alongside an adaptive time-EKF algorithm. Addressing robustness issues in conventional Kalman filtering, Zhang *et al.* (2023a) developed a joint vehicle state and parameter estimator using the maximum correlation entropy adaptive UKF. Heavy vehicle dynamics were also considered in Di Biase *et al.* (2020), where an EKF-based sideslip angle estimation method was designed using the Rational tire model. Meanwhile, Wei *et al.* (2022) formulated a dual EKF approach for estimating tire traction torque, and Wang *et al.* (2024) employed an expectation-maximization SRCKF algorithm for sideslip angle prediction. Expanding to mass and road slope estimation, Wang *et al.* (2022) combined multiple model recursive least squares (MMRLS) with Sage-Husa adaptive filtering and strong tracking filtering. Additionally, Zhang *et al.* (2024b) developed an ST-SRCKF observer to estimate vehicle state parameters and road adhesion coefficients, whereas Wang *et al.* (2023) introduced an adaptive UKF for vehicle state parameter estimation. Further contributions include a state parameter estimation technique leveraging a UKF with an ant lion algorithm (Liu & Cui, 2022) and a multi-dimensional information fusion-based parameter estimation method that integrates kinematic and dynamic Kalman filter observers (Tian *et al.*, 2022).

The data-driven approach, on the other hand, relies on existing state datasets, processing input–output relationships through intelligent algorithms such as neural networks, genetic algorithms, and fuzzy logic. In this realm, To optimize speed estimation in connected vehicles, Ji *et al.* (2023) introduced a consensus filtering algorithm for multi-source speed sensor networks, utilizing information weighting for improved accuracy. Machine learning techniques have also been widely adopted. For instance, Kumar *et al.* (2021) proposed an adaptive neural fuzzy system-based rotor flux estimator for induction motors, verifying its performance under varying loads. Meanwhile, Xu *et al.* (2021) explored a novel approach combining intelligent tires with machine learning to estimate tire slip angles with high precision. Optimization algorithms have also been utilized, as seen in You *et al.* (2024), where a vehicle state estimation algorithm based on the Whale Optimization Algorithm and Support Vector Regression (WOA-SVR) was formulated. Frequency response functions (FRF) were leveraged in Zhang *et al.* (2023b) to develop a vehicle parameter identification technique.

In recent years, hybrid methods that integrate physical models and data-driven technologies have become an important re-

search direction in the field of vehicle state estimation. Feng *et al.* (2023) systematically sorted out the main research directions of hybrid models, including changes in the order of using filtering algorithms and neural networks, training neural networks with filtering algorithms, and optimizing filtering algorithms with neural networks. Zhang *et al.* (2024a) devised a tire-road friction estimation method that integrates an enhanced CKF with an FNN, achieving superior accuracy in experimental validation. Zhou *et al.* (2023) employed the Levenberg–Marquardt backpropagation algorithm to estimate tire cornering stiffness, integrating it with a neural network-based time-varying Kalman filter (TVKF) for commercial vehicle sideslip angle estimation.

Existing state estimation methods based on vehicle physical models are mainly aimed at traditional non-articulated vehicles or based on a single dynamic model, while there are few studies on the strong nonlinear and multi-degree-of-freedom coupling characteristics of multi-axis distributed electric drive articulated vehicles. The existence of articulated devices significantly increases the complexity of the dynamic model of such vehicles, and the influence of the dynamic coupling of the front and rear bodies and the roll freedom degree must be considered at the same time. In addition, there are few comparative analyses of the impact of different degree-of-freedom models on estimation accuracy and the performance of filters under limited sensing conditions in existing work, resulting in a lack of theoretical basis for model and algorithm selection. Data-driven estimation methods and hybrid methods that integrate physical models and data-driven technologies have excellent nonlinear adaptability, but usually require a large amount of experimental data to train data-driven modules, with high computational complexity, and may be difficult to meet the real-time requirements of vehicle embedded systems. In contrast, this research method is based on physical models and nonlinear filtering theory, does not need to rely on historical data in specific scenarios, and avoids the problems of high data acquisition costs, annotation errors, and poor generalization of working conditions. EKF achieves efficient calculations through first-order linearization, and UKF balances nonlinear accuracy and computational burden through unscented transformation. Both have good real-time performance. This research method directly adapts to different working conditions through precise modeling and filtering algorithms, which can achieve accurate perception of vehicle status while reducing hardware costs. It is particularly suitable for the practical application of multi-axis distributed electric drive articulated vehicles in complex industrial scenarios.

In order to comprehensively describe the dynamic characteristics of the vehicle and study the impact of different dynamic models on state estimation, dynamic models including four degrees of freedom (4DOF) and six degrees of freedom (6DOF) were established. At the same time, EKF and UKF are the most mature nonlinear filtering methods in the field of vehicle state estimation. EKF has high computational efficiency and performs well under weak nonlinear conditions, but the linearization error is large. UKF improves the estimation accuracy through unscented transformation and is suitable for strong nonlinear conditions, but the computational complexity is slightly higher. The two complement each other and can comprehensively evaluate the effects of different dynamic models and filtering algorithms. Compared with previous studies, this study has made the following contributions:

- (i) Aiming at the strong nonlinearity and multi-degree-of-freedom coupling characteristics of multi-axis articulated vehicles, 4DOF and 6DOF dynamic models are proposed.

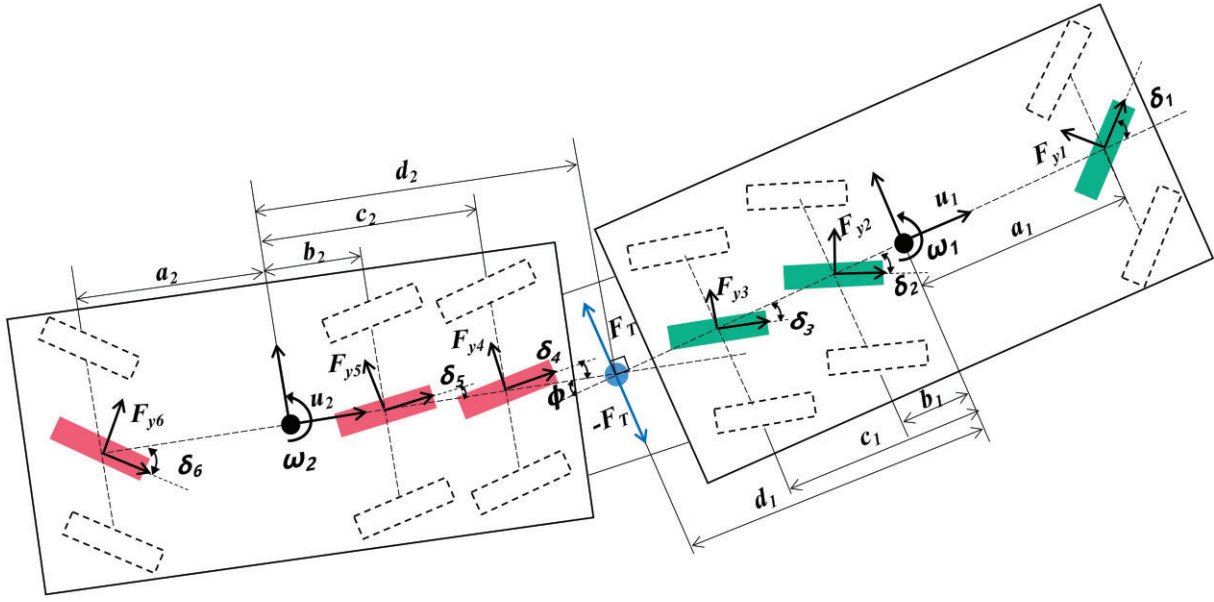


Figure 1: 4DOF vehicle model.

Among them, the 4DOF model focuses on the lateral-yaw motion coupling of the front and rear vehicle bodies, and the 6DOF model introduces the roll degree of freedom to improve the accuracy of high-speed conditions. Combining the EKF and UKF filtering frameworks, the EKF achieves efficient calculation of weak nonlinear scenarios through first-order Taylor linearization, and the UKF optimizes the robustness of strong nonlinear conditions based on unscented transformation, breaking through the limitations of traditional single model algorithms.

- (ii) An innovative model-algorithm matching criterion for state estimation of multi-axis articulated vehicles is proposed. In low-speed weak nonlinear scenarios, the 4DOF-EKF combination improves the computational efficiency when the error is controllable, and in high-speed strong nonlinear conditions, the 6DOF-UKF combination significantly improves the estimation accuracy. This strategy is verified by sine and fishhook simulations, providing a theoretical basis for the balance between computational efficiency and accuracy in different scenarios.
- (iii) The feasibility of the 6DOF-UKF solution was verified based on the Logitech G27 driving simulator, and millisecond-level synchronization between Trucksim and Simulink was achieved through the Real-Time Sync module. The research results can be directly applied to scenarios such as mines and ports, providing a high-precision, low-hardware-dependency state perception solution for articulated vehicle stability control and unmanned driving systems.

The remainder of this paper is organized as follows. Section 2 elaborates on the construction and verification of the 4DOF and 6DOF dynamic models of a multi-axis distributed electric drive articulated vehicle. Section 3 proposes a state estimator design framework based on EKF and UKF, and analyzes the applicability of the algorithm in nonlinear scenarios. Section 4 compares the estimation accuracy of different model-algorithm combinations through joint simulation tests of sinusoidal working conditions and fishhook working conditions, revealing the optimal matching strategy in low-speed weak nonlinear and high-speed strong non-

linear scenarios. Section 5 verifies the practical feasibility of the 6DOF-UKF combination based on the Logitech G27 driving simulator. Section 6 summarizes the research results and looks forward to the future direction of multi-model fusion and real-time optimization.

2. Vehicle Model

2.1. 4DOF vehicle model

The 4DOF model established by considering the lateral motion and yaw motion of unit 1 and the lateral motion and yaw motion of unit 2 has the following assumptions:

- (1) The longitudinal velocities of unit 1 and unit 2 are constant and equal.
- (2) Not considering the pitch and roll motion.
- (3) Each tire on each axis is assumed to be equivalent to a single tire.
- (4) Neglecting the influence of air resistance and road slope.

The simplified 4DOF model is shown in Figure 1.

Assuming that both δ_i and ϕ are small, then $\cos \delta_1 \approx 1$, $\cos \phi \approx 1$. The dynamics equations of the 4DOF vehicle model can be expressed as follows:

$$m_1 u_1 (\dot{\beta}_1 + \omega_1) = F_{y1} \cos \delta_1 + F_{y2} \cos \delta_2 + F_{y3} \cos \delta_3 + F_T, \quad (1)$$

$$I_{z1} \dot{\omega}_1 = a_1 F_{y1} \cos \delta_1 - b_1 F_{y2} \cos \delta_2 - c_1 F_{y3} \cos \delta_3 - d_1 F_T, \quad (2)$$

$$m_2 u_2 (\dot{\beta}_2 + \omega_2) = F_{y4} \cos \delta_4 + F_{y5} \cos \delta_5 + F_{y6} \cos \delta_6 - F_T \cos \phi, \quad (3)$$

$$I_{z2} \dot{\omega}_2 = -a_2 F_{y6} \cos \delta_6 + b_2 F_{y5} \cos \delta_5 + c_2 F_{y4} \cos \delta_4 - d_2 F_T \cos \phi. \quad (4)$$

In view of the low-speed operation characteristics of distributed electric drive articulated vehicles in industrial environments such as mines and ports, and considering that the actual application scenario is a high-friction road surface, the relationship between the tire lateral force and the sideslip angle can be accurately approximated by a linear model, so the linear tire model is selected. The total lateral forces acted on wheel can be obtained

as follows:

$$F_{yi} = k_i \alpha_i, \quad (5)$$

where k_i is the tire cornering stiffness, and the tire cornering stiffness parameters used are as follows: The first to sixth axles of the vehicle are $k_1 = -186578.7 \text{ N/rad}$, $k_2 = -179382.0 \text{ N/rad}$, $k_3 = -138377.2 \text{ N/rad}$, $k_4 = -186578.7 \text{ N/rad}$, $k_5 = -179382.0 \text{ N/rad}$, $k_6 = -138377.2 \text{ N/rad}$. α_i is the tire slip angle, and the calculation equation is as follows:

$$\begin{cases} \alpha_1 = \beta_1 + \frac{a_1 \omega_1}{u_1} - \delta_1 \\ \alpha_2 = \beta_1 - \frac{b_1 \omega_1}{u_1} - \delta_2 \\ \alpha_3 = \beta_1 - \frac{c_1 \omega_1}{u_1} - \delta_3 \\ \alpha_4 = \beta_2 + \frac{c_2 \omega_2}{u_2} - \delta_4 \\ \alpha_5 = \beta_2 + \frac{b_2 \omega_2}{u_2} - \delta_5 \\ \alpha_6 = \beta_2 - \frac{a_2 \omega_2}{u_2} - \delta_6 \end{cases} \quad (6)$$

Unit 1 and unit 2 are connected by a hinge, and the velocity relationship between them and the yaw angular velocity relationship are as follows:

$$\begin{aligned} u_2 &= u_1 \cos \phi - (v_1 - \omega_1 d_1) \sin \phi \\ v_2 &= u_1 \sin \phi + (v_1 - \omega_1 d_1) \cos \phi - \omega_2 d_2 \\ \dot{\phi} &= \omega_1 - \omega_2 \end{aligned} \quad (7)$$

Assuming that the articulation angle ϕ is small, the constraint equation between unit 1 and unit 2 can be obtained from Equation 7:

$$\dot{\beta}_1 - \dot{\beta}_2 - \frac{d_1}{u_1} \dot{\omega}_1 - \frac{d_2}{u_2} \dot{\omega}_2 + \omega_1 - \omega_2 = 0. \quad (8)$$

The following is the expression for the 4DOF model's state equation:

$$M\dot{X} = NX + PU, \quad (9)$$

where $X = [\beta_1, \omega_1, \beta_2, \omega_2]^T$, $U = [\delta_1, \delta_2, \delta_3, \delta_4, \delta_5, \delta_6]^T$,

$$\begin{aligned} M &= \begin{bmatrix} d_1 m_1 u_1 & I_{z1} & 0 & 0 \\ m_1 u_1 & 0 & m_2 u_2 & 0 \\ 0 & 0 & d_2 m_2 u_2 & -I_{z2} \\ -1 & -\frac{d_1}{u_1} & 1 & -\frac{d_2}{u_2} \end{bmatrix}, \\ N &= \begin{bmatrix} N_{11} & N_{12} & 0 & 0 \\ N_{21} & N_{22} & N_{23} & N_{24} \\ 0 & 0 & N_{33} & N_{34} \\ 0 & -1 & 0 & 1 \end{bmatrix}, \\ N_{11} &= ak_1 - bk_2 - ck_3 + d(k_1 + k_2 + k_3), \\ N_{12} &= \frac{k_1(ad + a^2) - k_2(bd - b^2) - k_3(cd - c^2)}{u_1} - m_1 d_1 u_1, \\ N_{21} &= k_1 + k_2 + k_3, N_{22} = \frac{ak_1 - bk_2 - ck_3}{u_1} - m_1 u_1, \\ N_{23} &= k_4 + k_5 + k_6, N_{24} = \frac{ck_4 + bk_5 - ak_6}{u_2} - m_2 u_2, \\ N_{33} &= ak_6 - bk_5 - ck_4 + d(k_4 + k_5 + k_6), \\ N_{34} &= \frac{k_4(cd - c^2) + k_5(bd - d^2) - k_6(ad + a^2)}{u_2} - m_2 d_2 u_2, \\ P &= \begin{bmatrix} P_{11} & P_{12} & P_{13} & 0 & 0 & 0 \\ -k_1 & -k_2 & -k_3 & -k_4 & -k_5 & -k_6 \\ 0 & 0 & 0 & P_{34} & (b-d)k_5 & -(a+d)k_6 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \end{aligned}$$

where $P_{11} = -(a+d)k_1$, $P_{12} = (b-d)k_2$, $P_{13} = (c-d)k_3$, $P_{34} = (c-d)k_4$.

The space equation of state for the 4DOF model can be expressed as follows:

$$\dot{X} = AX + BU, \quad (10)$$

where $A = M^{-1}N$ and $B = M^{-1}P$.

2.2. 6DOF vehicle model

This study adds a degree of freedom for vehicle roll on the basis of the original 4DOF, and establishes a 6DOF vehicle model. Considering the lateral motion, yaw motion, and roll motion of unit 1, and the lateral motion, yaw motion, and roll motion of unit 2, the established roll motion model is shown in Figure 2.

In Figure 2, m_{s1} , m_{s2} are the sprung masses of unit 1 and unit 2, h_1 , h_2 are the distances from the center of mass of the sprung masses of unit 1 and unit 2 to the roll center, h_{1c} , h_{2c} are the distances from the hinge to the roll center of unit 1 and unit 2, φ_1 , φ_2 are the roll angles of unit 1 and unit 2, and O_1 , O_2 are the roll centers of unit 1 and unit 2.

The following is an expression for the 6DOF model's dynamics equations:

$$m_1 u_1 (\dot{\beta}_1 + \omega_1) - m_{s1} h_1 \ddot{\varphi}_1 = F_{y1} \cos \delta_1 + F_{y2} \cos \delta_2 + F_{y3} \cos \delta_3 + F_T, \quad (11)$$

$$I_{z1} \dot{\omega}_1 - I_{1xz} \ddot{\varphi}_1 = a_1 F_{y1} \cos \delta_1 - b_1 F_{y2} \cos \delta_2 - c_1 F_{y3} \cos \delta_3 - d_1 F_T, \quad (12)$$

$$\begin{aligned} (I_{1xx} + m_{s1} h_1^2) \ddot{\varphi}_1 - I_{1xz} \dot{\omega}_1 &= m_{s1} h_1 [u_1 (\dot{\beta}_1 + \omega_1) - h_1 \ddot{\varphi}_1] \\ &+ m_{s1} g h_1 \varphi_1 - K_1 \varphi_1 - C_1 \dot{\varphi}_1 + F_T h_{1c}, \end{aligned} \quad (13)$$

$$m_2 u_2 (\dot{\beta}_2 + \omega_2) - m_{s2} h_2 \ddot{\varphi}_2 = F_{y4} \cos \delta_4 + F_{y5} \cos \delta_5 + F_{y6} \cos \delta_6 - F_T, \quad (14)$$

$$\begin{aligned} I_{z2} \dot{\omega}_2 - I_{2xz} \ddot{\varphi}_2 &= c_2 F_{y4} \cos \delta_4 + b_2 F_{y5} \cos \delta_5 - a_2 F_{y6} \cos \delta_6 - d_2 F_T, \\ (I_{2xx} + m_{s2} h_2^2) \ddot{\varphi}_2 - I_{2xz} \dot{\omega}_2 &= m_{s2} h_2 [u_2 (\dot{\beta}_2 + \omega_2) - h_2 \ddot{\varphi}_2] \\ &+ m_{s2} g h_2 \varphi_2 - K_2 \varphi_2 - C_2 \dot{\varphi}_2 - F_T h_{2c}. \end{aligned} \quad (15)$$

The constraint equation for unit 1 and unit 2 is expressed as follows:

$$\dot{\beta}_1 - \dot{\beta}_2 - \frac{h_{1c}}{u_1} \ddot{\varphi}_1 + \frac{h_{2c}}{u_2} \ddot{\varphi}_2 - \frac{d_1}{u_1} \dot{\omega}_1 - \frac{d_2}{u_2} \dot{\omega}_2 + \omega_1 - \omega_2 = 0. \quad (17)$$

The tire model is the same as the previous text and will not be repeated here. The following is the expression for the 6DOF model's state equation:

$$\dot{X} = \bar{A}X + \bar{B}U, \quad (18)$$

where $X = [\beta_1, \omega_1, \varphi_1, \dot{\varphi}_1, \beta_2, \omega_2, \varphi_2, \dot{\varphi}_2]^T$, $\bar{A} = \bar{M}^{-1}\bar{N}$, $\bar{B} = \bar{M}^{-1}\bar{P}$, $\bar{M}$, $\bar{N}$, and $\bar{P}$ are provided in Appendix A.

2.3. Ackermann steering model

The Ackermann steering principle ensures that the wheels do not slip when the vehicle is steering and are in a pure rolling state, which reduces tire wear, decreases the turning radius of the vehicle, and improves the trafficability. The distributed electric drive articulated vehicle studied in this paper is all-wheel steering, and the established Ackermann steering model is demonstrated in Figure 3.

According to the Ackermann steering principle and geometric relationship, the front wheel angle and articulation angle are provided by Trucksim. Using the small angle assumption, the calcu-

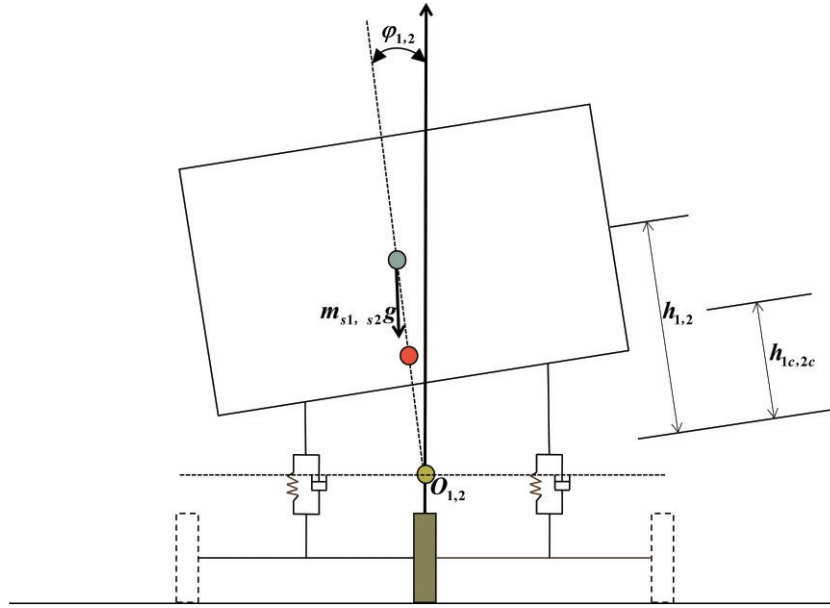


Figure 2: Schematic diagram of roll motion model.

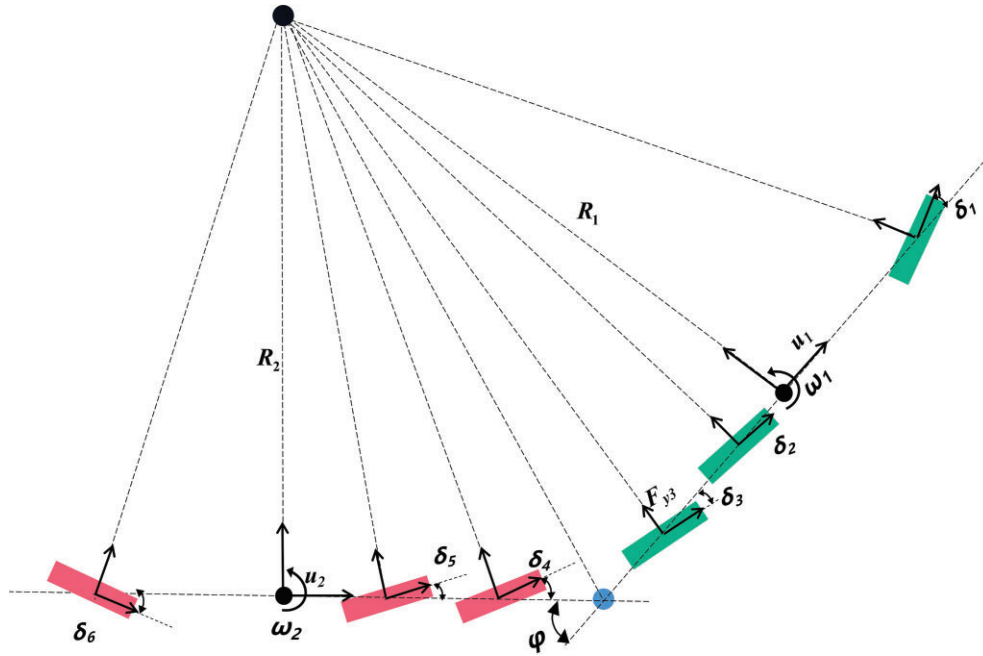


Figure 3: Ackermann steering model.

lated wheel angles for each axle are as follows:

$$\begin{cases} \tan \delta_1 = \frac{a_1}{R_1} \\ \tan \delta_2 = \frac{b_1}{R_1} \\ \tan \delta_3 = \frac{c_1}{R_1} \\ \tan \frac{\phi}{2} = \frac{d_2}{R_2} \\ \tan \delta_4 = \frac{c_2}{R_2} \\ \tan \delta_5 = \frac{b_2}{R_2} \\ \tan \delta_6 = \frac{a_2}{R_2} \end{cases} \rightarrow \begin{cases} \delta_2 \approx \tan \delta_2 = \frac{b_1}{a_1} \delta_1 \\ \delta_3 \approx \tan \delta_3 = \frac{c_1}{a_1} \delta_1 \\ \delta_4 \approx \tan \delta_4 = \frac{c_2 \phi}{2d_2} \\ \delta_5 \approx \tan \delta_5 = \frac{b_2 \phi}{2d_2} \\ \delta_6 \approx \tan \delta_6 = \frac{a_2 \phi}{2d_2} \end{cases} \quad (19)$$

2.4. Motor model

The focus of this study is on state estimation of distributed electric drive articulated vehicle. Therefore the motor model is sim-

plified. The torque characteristic curve model of the permanent magnet synchronous motor can be expressed as

$$T_i = \begin{cases} T_e & n < n_e \\ \frac{9549P_e}{n} & n > n_e \end{cases}, \quad (20)$$

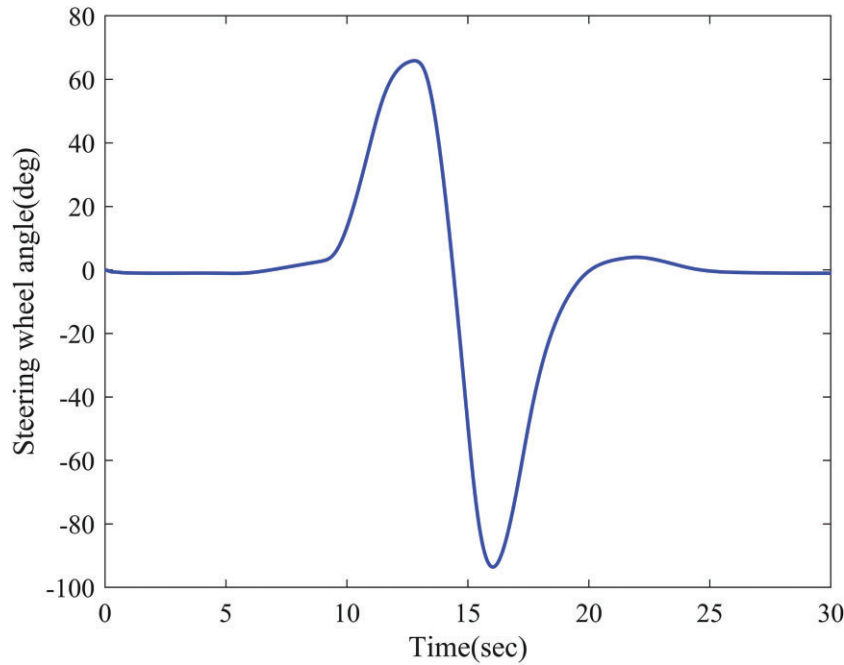
where T_i ($i = 1, 2, 3, 4, 5, 6$) is the output torque of the motor for the i th axle. The rated power of the motor is denoted by P_e , the rated output torque by T_e , the actual speed by n , and the base speed by n_e .

2.5. Dynamics model validation

In order to verify the accuracy of the above constructed model, a distributed electric drive articulated vehicle model was first

Table 1: Vehicle parameters.

Symbol	Meaning	Value	Symbol	Meaning	Value
m_1	mass of the unit 1	5505 kg	m_2	mass of the unit 2	6975 kg
a_1	Distance from CG of the unit 1 to front axle	4.134 m	I_{1xx}	Inertia moment of mass about the longitudinal axis of unit 1	2286.8 kg · m ²
a_2	Distance from CG of the unit 2 to fourth axle	4.134 m	I_{2xx}	Inertia moment of mass about the longitudinal axis of unit 2	9959.7 kg · m ²
b_1	Distance from CG of the unit 1 to second axle	1.483 m	I_{1xz}	Lateral roll inertia product of the unit 1 around the CG	1626 kg · m ²
b_2	Distance from CG of the unit 2 to fifth axle	1.483 m	I_{2xz}	Lateral roll inertia product of the unit 2 around the CG	0
c_1	Distance from CG of the unit 1 to third axle	3.133 m	K_1	Roll stiffness of unit 1	343 775 N · m · deg ⁻¹
c_2	Distance from CG of the unit 2 to sixth axle	3.133 m	K_2	Roll stiffness of unit 2	171 887 N · m · deg ⁻¹
d_1	Distance from CG of the unit 1 to hinge joint	4.858 m	C_1	Roll damping of unit 1	57 296
d_2	Distance from CG of the unit 2 to hinge joint	4.858 m	C_2	Roll damping of unit 1	51 566
I_{z1}	Inertia moment of mass about the vertical axis of unit 1	34 823.2 kg · m ²	h_1	Distance from the CG of the unit 1 to the center of roll	1.175 m
I_{z2}	Inertia moment of mass about the vertical axis of unit 2	179 992 kg · m ²	h_2	Distance from the CG of the unit 2 to the center of roll	1m
m_{s1}	Sprung mass of unit 1	4455 kg	h_{1c}	Distance from the hinge joint to the roll center of the unit 1	0.572 m
m_{s2}	Sprung mass of unit 2	5925 kg	h_{2c}	Distance from the hinge joint to the roll center of the unit 2	0.572 m

**Figure 4:** Steering wheel angle.

established using Truksim and MATLAB/Simulink as a reference model. Then, the 4DOF and 6DOF model are simulated and compared with the reference model, and the effectiveness of the model was verified by comparing the vehicle's state variables. The vehicle parameters are shown in Table 1.

The simulation condition is selected as the single lane change condition, with a vehicle speed of 20 km/h and a road adhesion coefficient of 0.85. The steering wheel angle input is shown in Figure 4. The 4DOF vehicle model and the 6DOF vehicle model corresponding simulation verification results are shown in Figures 5 and 6.

According to the verification results, the 4DOF model has a high degree of fit with the reference model in terms of response trend and response amplitude for the yaw rate of unit 1 and unit 2. The overall trend of the sideslip angle of unit 1 and unit 2 is similar, but the response trend is slightly delayed and there is a deviation in the response amplitude. This is because the vehicle model studied in this paper is relatively long and has a large wheelbase. And the 4DOF model ignores the influence of roll motion and suspension system, resulting in deviation. However, the deviation is within a reasonable range, and the overall effect is relatively accurate. The 6DOF model has a higher fitting degree in response trend and response amplitude for the yaw rate of unit 1 and unit 2, and

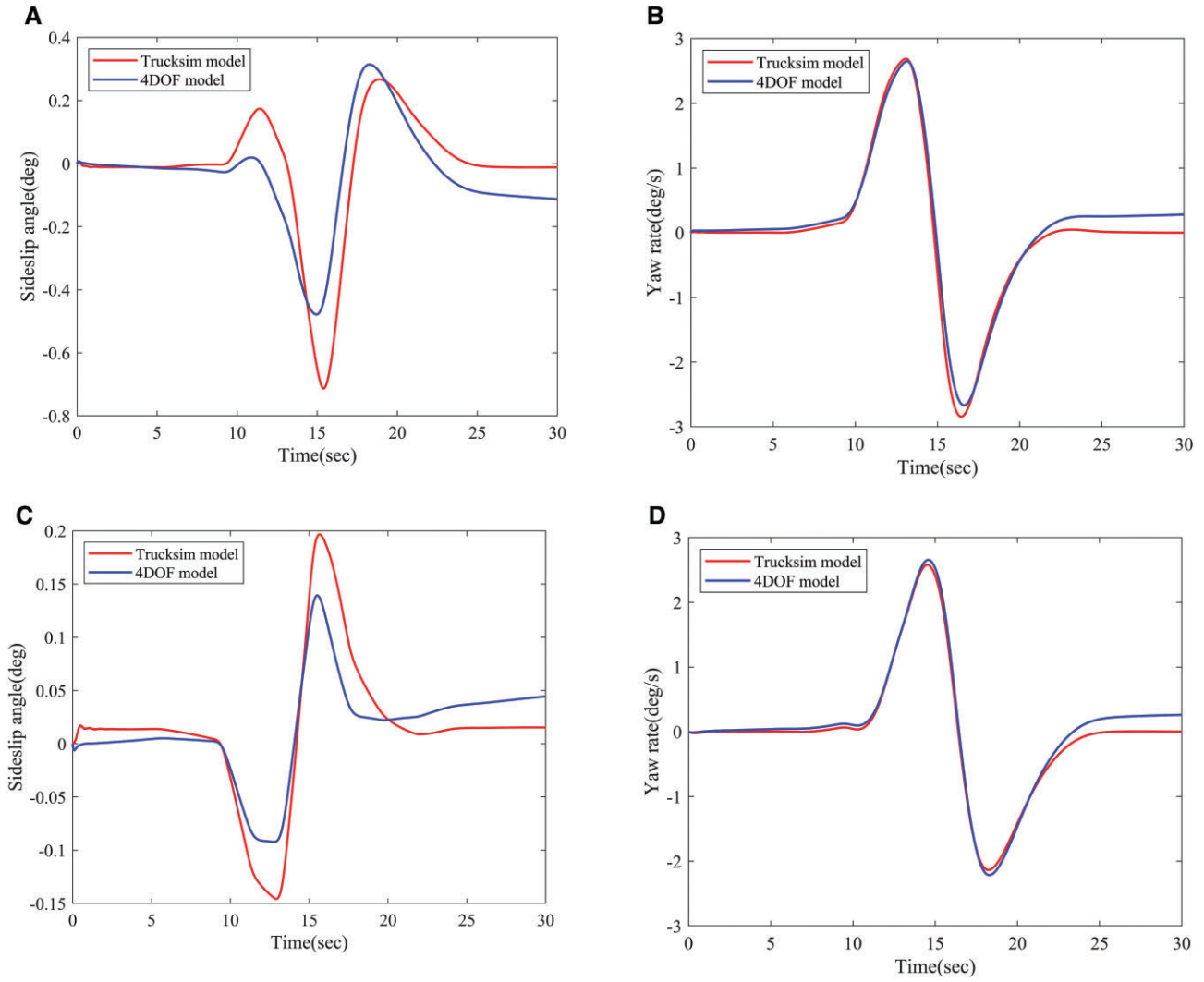


Figure 5: Comparison between 4DOF model and reference model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

a better response trend and response amplitude for the sideslip angle of unit 1 and unit 2. This is because the 6DOF model considers the influence of roll motion, and the accuracy of the response results is higher than that of the 4DOF model. Therefore, the 4DOF and 6DOF models constructed have high accuracy, can comprehensively reflect the dynamic response characteristics of the real vehicle, and provide a model basis for vehicle state estimation research.

3. State Estimator Based on Nonlinear Kalman Filter

3.1. Equation of state and observation equation

The state equation of the 4DOF vehicle model established in the previous text is

$$\dot{X} = AX + BU,$$

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}, B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} & b_{15} & b_{16} \\ b_{21} & b_{22} & b_{23} & b_{24} & b_{25} & b_{26} \\ b_{31} & b_{32} & b_{33} & b_{34} & b_{35} & b_{36} \\ b_{41} & b_{42} & b_{43} & b_{44} & b_{45} & b_{46} \end{bmatrix}.$$

As can be seen from the above equation:

$$\begin{aligned} \dot{\beta}_1 = & a_{11}\beta_1 + a_{12}\omega_1 + a_{13}\beta_2 + a_{14}\omega_2 + b_{11}\delta_1 + b_{12}\delta_2 + b_{13}\delta_3 + b_{14}\delta_4 \\ & + b_{15}\delta_5 + b_{16}\delta_6. \end{aligned} \quad (21)$$

The lateral acceleration of the unit 1 is: $a_{y1} = u_1(\dot{\beta}_1 + \omega_1)$.

Combining the above equations yields the following equation:

$$\begin{aligned} a_{y1} = CX + DU = u_1 & \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{12} + 1 & 0 & 0 \\ 0 & 0 & a_{13} & 0 \\ 0 & 0 & 0 & a_{14} \end{bmatrix} \begin{bmatrix} \beta_1 \\ \omega_1 \\ \beta_2 \\ \omega_2 \end{bmatrix} \\ & \times \begin{bmatrix} b_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & b_{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & b_{13} & 0 & 0 & 0 \\ 0 & 0 & 0 & b_{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & b_{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & b_{16} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \\ \delta_5 \\ \delta_6 \end{bmatrix}, \end{aligned} \quad (22)$$

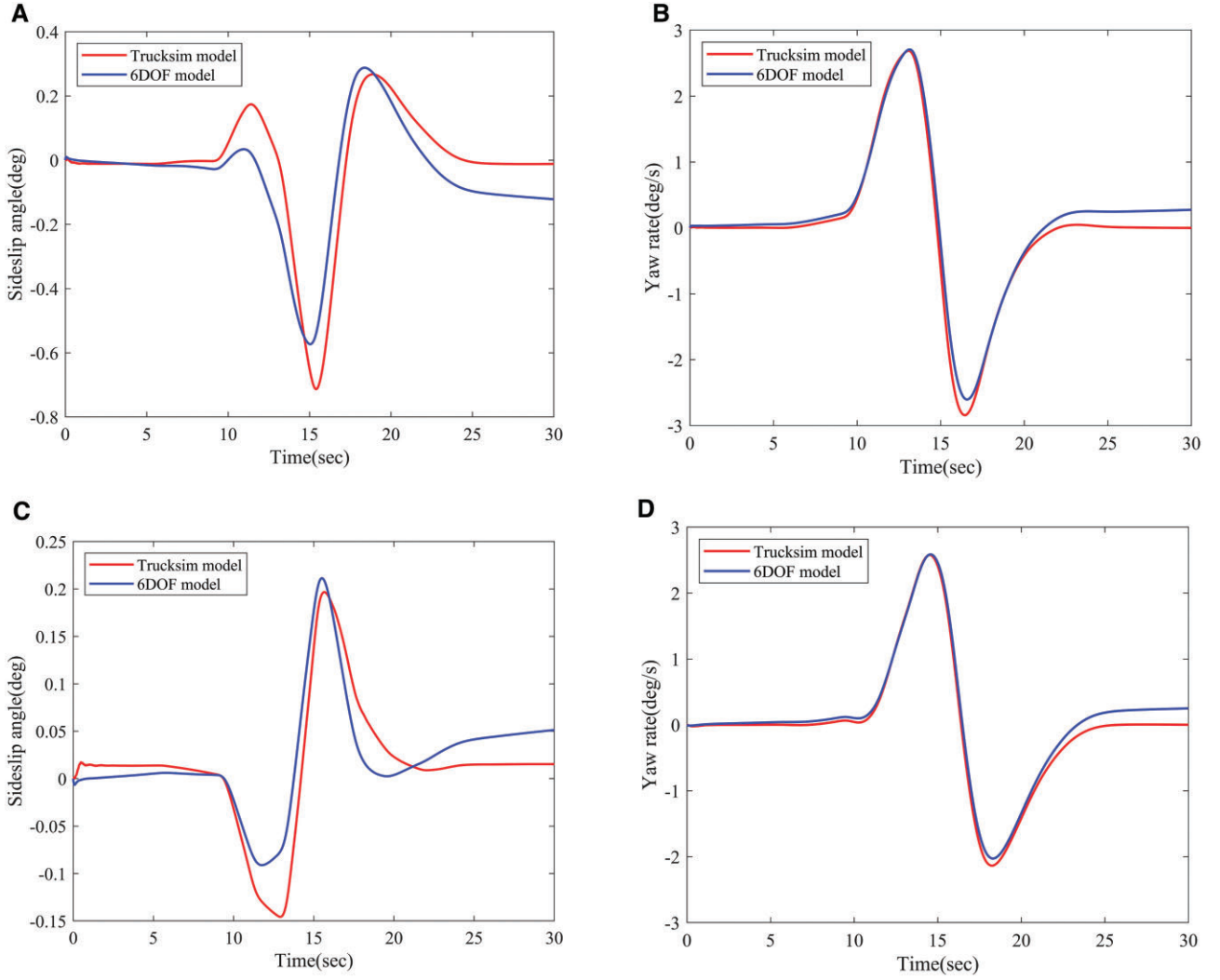


Figure 6: Comparison between 6DOF model and reference model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

where

$$C = u_1 \begin{bmatrix} a_{11} & 0 & 0 & 0 \\ 0 & a_{12} + 1 & 0 & 0 \\ 0 & 0 & a_{13} & 0 \\ 0 & 0 & 0 & a_{14} \end{bmatrix}, \quad D =$$

$$u_1 \begin{bmatrix} b_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & b_{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & b_{13} & 0 & 0 & 0 \\ 0 & 0 & 0 & b_{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & b_{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & b_{16} \end{bmatrix}.$$

Based on the 4DOF model established above and the rewritten equation for the lateral acceleration of the unit 1, the normalized equations of state and observation equations can be obtained:

$$\begin{cases} \dot{X}(t) = f(x(t), u(t), w(t)) \\ Z(t) = h(x(t), u(t), v(t)) \end{cases}, \quad (23)$$

where $w(t)$ and $v(t)$ are the process noise and observation noise with covariance matrices Q and R , respectively. $x(t) = [\beta_1, \omega_1, \beta_2, \omega_2]^T$, $u(t) = [\delta_1, \delta_2, \delta_3, \delta_4, \delta_5, \delta_6]^T$, $z(t) = [a_{y1}]$.

The equations of state and observation equations for the 6DOF model can be established according to the above method, which will not be repeated here.

3.2. EKF state estimator

The EKF performs first-order linearization of the nonlinear system through Taylor expansion, avoiding the direct solution of complex nonlinear problems, and has the characteristics of high computational efficiency and simple implementation. The dynamic model of distributed electric drive articulated vehicles has highly nonlinear and multi-degree-of-freedom coupling characteristics and is suitable for low-speed weak nonlinear conditions. Under high-speed strong nonlinear conditions, the system nonlinearity is enhanced and the error may be large, resulting in a decrease in estimation accuracy. The following is the EKF's computational process:

$$\begin{cases} A = \frac{\partial f}{\partial x} \\ H = \frac{\partial z}{\partial x} \end{cases}, \quad (24)$$

where A is the Jacobian matrix of the nonlinear function f for the partial derivation of the state quantity x , and H is the Jacobian matrix of the nonlinear function h for the partial derivation of the state quantity x .

First, the initial values of the state vector and state error covariance are set to $\hat{x}_0$ and p_0 , and the time update is started for

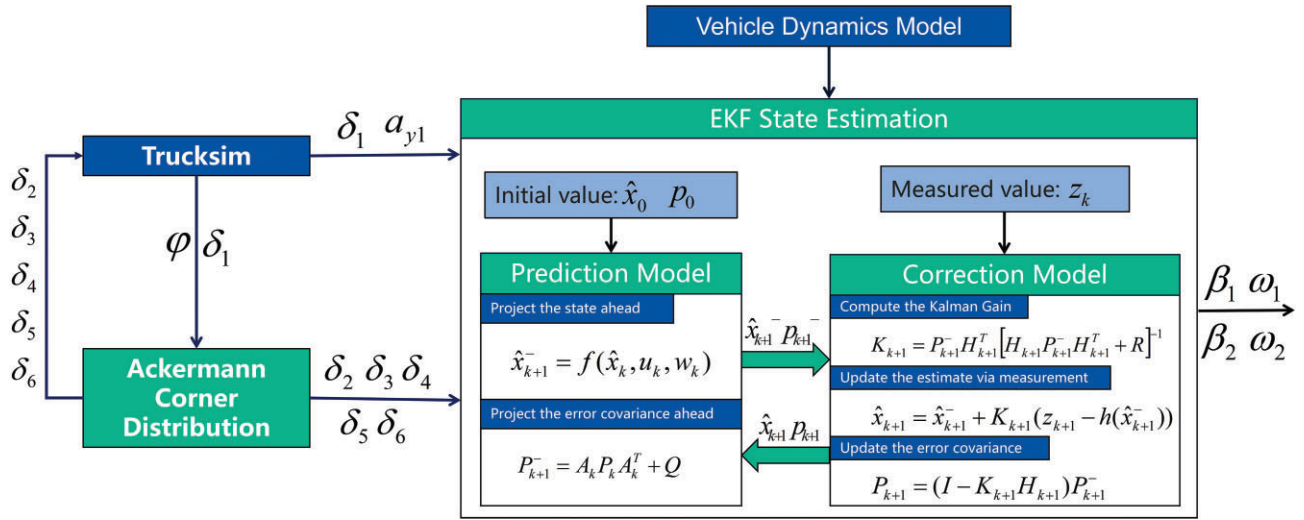


Figure 7: Schematic diagram of EKF vehicle state estimator.

state prediction:

$$\hat{x}_{k+1}^- = f(\hat{x}_k, u_k, w_k). \quad (25)$$

The second step of performing the time update, state error covariance prediction:

$$P_{k+1}^- = A_k P_k A_k^T + Q. \quad (26)$$

Start the measurement update process for Kalman gain matrix calculations:

$$K_{k+1} = \frac{P_{k+1}^- H_{k+1}^T}{H_{k+1} P_{k+1}^- H_{k+1}^T + R}. \quad (27)$$

The second step of the measurement update process is carried out, where the state is correctively updated by the accessible observations:

$$\hat{x}_{k+1} = \hat{x}_{k+1}^- + K_{k+1} (z_{k+1} - h(\hat{x}_{k+1}^-)), \quad (28)$$

The third step of the measurement update process is performed to correct the state error covariance for updating:

$$P_{k+1} = (I - K_{k+1} H_{k+1}) P_{k+1}^-. \quad (29)$$

According to the above steps of the EKF state estimation algorithm, an EKF vehicle state estimator as shown in Figure 7 was established. The EKF state estimator receives unit 1's lateral acceleration and the front wheel angle from the Trucksim model, while the Ackermann steering model sends additional input, namely the angles of the other axes. The 4DOF model or 6DOF model established in the schematic diagram can be used to study the impact of different dynamic model on the vehicle state estimation results under the same state estimation algorithm.

3.3. UKF state estimator

The UKF uses unscented transformation to directly approximate the probability distribution of nonlinear systems, without calculating the Jacobian matrix, avoiding linearization errors, and is particularly suitable for strongly nonlinear systems. In the state estimation of distributed electric drive articulated vehicles, UKF can more accurately retain nonlinear characteristics, but UKF needs to generate and propagate multiple Sigma points, and the amount of calculation is significantly higher than EKF. If the degree of freedom of the vehicle model increases, the expansion of

the state dimension will further increase the computational burden.

Firstly, set the initial values for the state variables and covariance:

$$\hat{x}_0 = E(x_0), \quad P_0 = E[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T]. \quad (30)$$

Using a symmetric sampling strategy to construct a Sigma sampling point set χ_i and obtain $2n + 1$ sigma sampling points:

$$\chi_0 = \bar{x}, \quad i = 0,$$

$$\chi_i = \bar{x} + (\sqrt{(n + \lambda) P_x})_i, \quad i = 1, 2, \dots, n,$$

$$\chi_i = \bar{x} - (\sqrt{(n + \lambda) P_x})_{i-n}, \quad i = n + 1, n + 2, \dots, 2n,$$

where n is the dimension of the state vector, $\bar{x}$ and P_x are the mean and covariance matrix, respectively. $\lambda > 0$ is the scaling factor, satisfying $\lambda = \alpha^2(n + \kappa) - n$. α is the range of Sigma point set near the mean $\bar{x}$, usually taken as a small value. κ is the second-order scaling factor, taken as $\kappa = 0$.

Calculate the weight of sampling points:

$$\omega_0^m = \frac{\lambda}{n + \lambda}, \quad i = 0,$$

$$\omega_0^c = \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta), \quad i = 0,$$

$$\omega_i^m = \omega_i^c = \frac{\lambda}{2(n + \lambda)}, \quad i = 1, 2, \dots, 2n,$$

where the superscript m represents the corresponding weight of the mean, the superscript c represents the corresponding weight of the covariance, $\beta \geq 0$ is the weight coefficient. Assuming that the system noise is approximately Gaussian distributed, setting β to 2 can give more reasonable weights to statistical features other than the mean and covariance, thereby improving filtering accuracy, so $\beta = 2$ is taken.

Next is the time update section, which mainly includes state quantity prediction and observation quantity prediction. Based on the constructed Sigma sampling point set, obtain the corresponding mapping point set:

$$\chi_{i,k+1} = f(\chi_i), \quad (31)$$

Perform state quantity prediction and calculate the prior estimate and prior state error covariance of the state quantity at time $k + 1$:

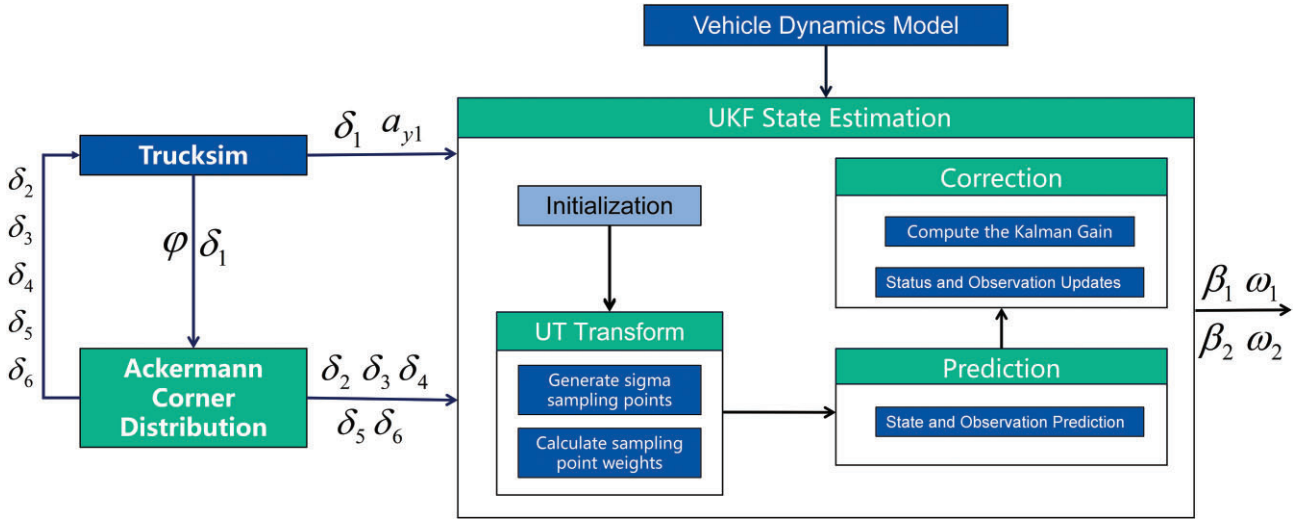


Figure 8: Schematic diagram of UKF vehicle state estimator.

$$\hat{x}_{k+1}^- = \sum_{i=0}^{2n} \omega_i^m x_{i,k+1}, \quad (32)$$

$$P_{x,k+1}^- = \sum_{i=0}^{2n} \omega_i^c (x_{i,k+1} - \hat{x}_{k+1}^-) (x_{i,k+1} - \hat{x}_{k+1}^-)^T + Q, \quad (33)$$

Perform observation prediction and calculate the estimated value and covariance of the observation at time $k+1$:

$$\hat{z}_{k+1}^- = \sum_{i=0}^{2n} \omega_i^m z_{i,k+1}, \quad (34)$$

$$P_{z,k+1}^- = \sum_{i=0}^{2n} \omega_i^c (z_{i,k+1} - \hat{z}_{k+1}^-) (z_{i,k+1} - \hat{z}_{k+1}^-)^T + R. \quad (35)$$

Calculate the covariance matrix between the state quantities and the observed quantities:

$$P_{xz,k+1}^- = \sum_{i=0}^{2n} \omega_i^c (x_{i,k+1} - \hat{x}_{k+1}^-) (z_{i,k+1} - \hat{z}_{k+1}^-)^T. \quad (36)$$

The correction update part is carried out below to calculate the Kalman gain:

$$K_{k+1} = P_{xz,k+1}^- P_{z,k+1}^{-1}, \quad (37)$$

Perform correction and update of state variables, calculate posterior estimates of state variables:

$$\hat{x}_{k+1} = \hat{x}_{k+1}^- + K_{k+1} (z_{k+1} - \hat{z}_{k+1}^-). \quad (38)$$

Perform correction and update of state error covariance, calculate posterior state error covariance:

$$P_{x,k+1} = P_{x,k+1}^- - K_{k+1} P_{z,k+1} K_{k+1}^T. \quad (39)$$

Based on the above steps of the UKF state estimation algorithm, a UKF vehicle state estimator as shown in Figure 8 is established. When designing the UKF vehicle state estimator, except for using different estimation algorithms, all other modules are consistent with the EKF vehicle state estimation schematic diagram.

4. Simulation Results

In order to achieve accurate estimation of distributed electric drive articulated vehicle, two estimation methods and two vehicle dynamics models are analyzed and compared. Simulation experiments are conducted using the Trucksim and Simulink joint

simulation platform, and sinusoidal and fishhook conditions are selected as the test conditions. Due to the special body structure of distributed electric drive articulated vehicle, the driving speed is relatively low in practical application scenarios, so 20 and 40 km/h are selected as the vehicle speeds.

The initial values of the state variables for EKF state estimator and UKF state estimator are $[0, 0, 0, 0]^T$. The initial value of the covariance matrix for the 4DOF model is $Q = \text{diag}[0.001, 0.001, 0.001, 0.001]^T$, the initial value of the covariance matrix for the 6DOF model is $Q = \text{diag}[0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 0.001]^T$. The initial value of the observation noise covariance is $R = 1$, the sampling time is 0.001s. In order to verify the robustness of the algorithm to nonlinear interference and avoid signal distortion due to excessively high noise amplitude or signal deviating from the actual engineering boundary due to too low noise amplitude, random noise with a mean of 0 and a variance of 0.1 is superimposed on the observed value to simulate the sensor measurement noise.

4.1. Sinusoidal condition

The sine input condition for the front wheel steering angle is shown in Figure 9, with a road adhesion coefficient of 0.85 and a simulation time of 12s. The simulation results based on the 4DOF model are shown in Figure 10, and the error plots are shown in Figure 11. The simulation results based on the 6DOF model are shown in Figure 12, and the error plots are shown in Figure 13.

In order to compare the estimation accuracy, the mean absolute error (MAE) and root mean square error (RMSE) are calculated for 20 and 40 km/h, respectively, as shown in Tables 2 and 3. The calculation expressions for MAE and RMSE are as follows:

$$\text{MAE} = \frac{\sum_{i=1}^N |\hat{x}_i - x_i|}{N}, \quad (40)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^N (\hat{x}_i - x_i)^2}{N}}, \quad (41)$$

where $\hat{x}_i$, x_i are the estimated value and reference value, respectively, N is the total number of samples.

The calculation results show that under the sine working condition, when comparing the results of EKF and UKF, in the estimation of the sideslip angle of the center of mass, whether it is 20 or

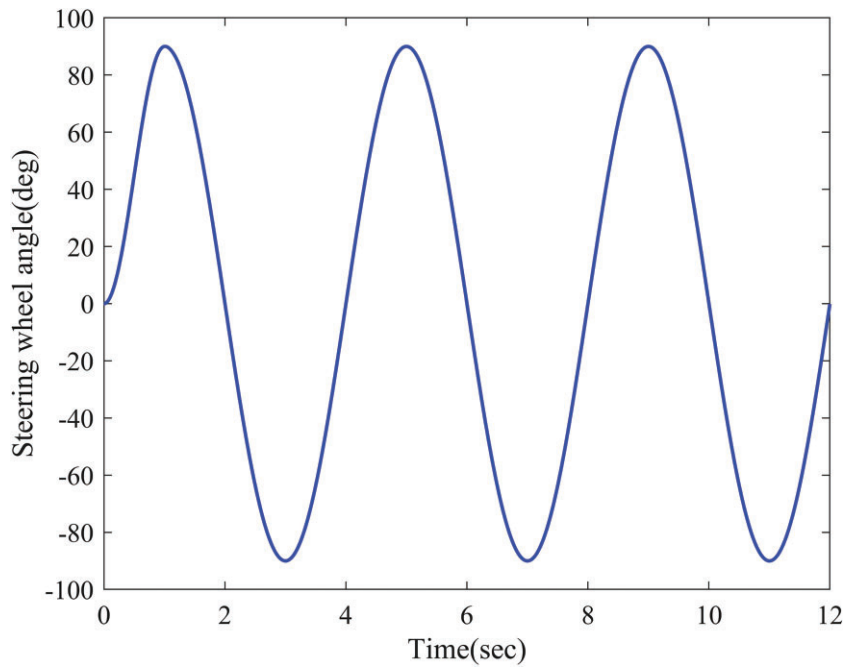


Figure 9: Steering wheel angle.

40 km/h, UKF only has an advantage in some indicators, and the estimation accuracy of the two is roughly the same. In the estimation of the yaw rate, UKF is better than EKF, with a maximum reduction of 20.4% in MAE and a maximum reduction of 18% in RMSE. When comparing the results of 4DOF and 6DOF, in the estimation of the sideslip angle of the center of mass, at 20 km/h, EKF under 6DOF is better than EKF under 4DOF, with a maximum reduction of 54.3% in MAE and a maximum reduction of 50% in RMSE, while the accuracy of UKF under 6DOF is close to that of UKF under 4DOF. At 40 km/h, 4DOF is better than 6DOF, with a maximum reduction of 29% in MAE and a maximum reduction of 31.7% in RMSE. In the estimation of yaw angular velocity, at 20 km/h, 4DOF is better than 6DOF, with MAE reduced by 38.1% and RMSE reduced by 35.2% at most. At 40 km/h, 6DOF is better than 4DOF, with MAE reduced by 37.2% and RMSE reduced by 35.0% at most.

4.2. Fishhook condition

The steering wheel angle input for the fishhook condition is shown in Figure 14, with a road adhesion coefficient of 0.85 and a simulation time of 12 seconds. The simulation results based on the 4DOF model are shown in Figure 15, and the error plots are shown in Figure 16. The simulation results based on the 6DOF model are shown in Figure 17, and the error plots are shown in Figure 18. The MAE and RMSE were calculated at speeds of 20 km/h and 40 km/h, as shown in Table 4 and Table 5.

From the above calculation results, it can be seen that under the fishhook condition, when comparing the results of EKF and UKF, in the estimation of the sideslip angle of the center of mass, whether it is 20 or 40 km/h, UKF only has an advantage in some indicators, and the estimation accuracy of the two is roughly the same. In the estimation of yaw angular velocity, UKF is better than EKF, with a maximum reduction of 64% in MAE and a maximum reduction of 64.9% in RMSE. Comparing the results of 4DOF and 6DOF, in the estimation of the sideslip angle of the center of mass, at 20 km/h, 6DOF is better than 4DOF, with a maximum reduction

of 62.6% in MAE and a maximum reduction of 62.8% in RMSE. At 40 km/h, 6DOF is better than 4DOF, with a maximum reduction of 37.5% in MAE and a maximum reduction of 26.6% in RMSE. In the estimation of yaw rate, at 20 km/h, 4DOF is better than 6DOF, MAE is reduced by 65.6% at most, RMSE is reduced by 62.7% at most, at 40 km/h, 6DOF is better than 4DOF, MAE is reduced by 48.4% at most, RMSE is reduced by 48.7% at most.

According to the simulation results above, EKF is comparable to UKF in estimating the sideslip angle of the center of mass. EKF performs better in some aspects because the dynamic characteristics of the sideslip angle of the center of mass are directly affected by the lateral force of the vehicle and the linear model of the tire, and the approximate linear characteristics of the system dominate. At this time, the linearization method of EKF through the first-order Taylor expansion is sufficient to effectively describe the system. Although the unscented transformation of UKF is theoretically better, the computational complexity may introduce a small difference in noise sensitivity, resulting in the two accuracies being close or even inferior to the estimation accuracy of EKF. In the estimation of yaw rate, UKF is worse than EKF because the dynamic characteristics of yaw rate involve multi-degree-of-freedom coupling and have strong nonlinear characteristics. UKF directly approximates the nonlinear probability distribution through unscented transformation, and more accurately captures the dynamic transmission of torque in the vehicle model, while the linearization of EKF underestimates such coupling effects.

According to the comparison of the models, in terms of the estimation of the sideslip angle of the center of mass, 6DOF is generally better than 4DOF. This is because the 6DOF model introduces an additional roll degree of freedom, which can more realistically reflect the dynamic state of the vehicle. In terms of the estimation of the yaw rate, the two are generally equally competitive. This is because the yaw rate is mainly controlled by the yaw moment balance. Although the 4DOF model ignores the roll degree of freedom, its simplified dynamic equation may indirectly compensate for the roll effect through parameter equivalence,

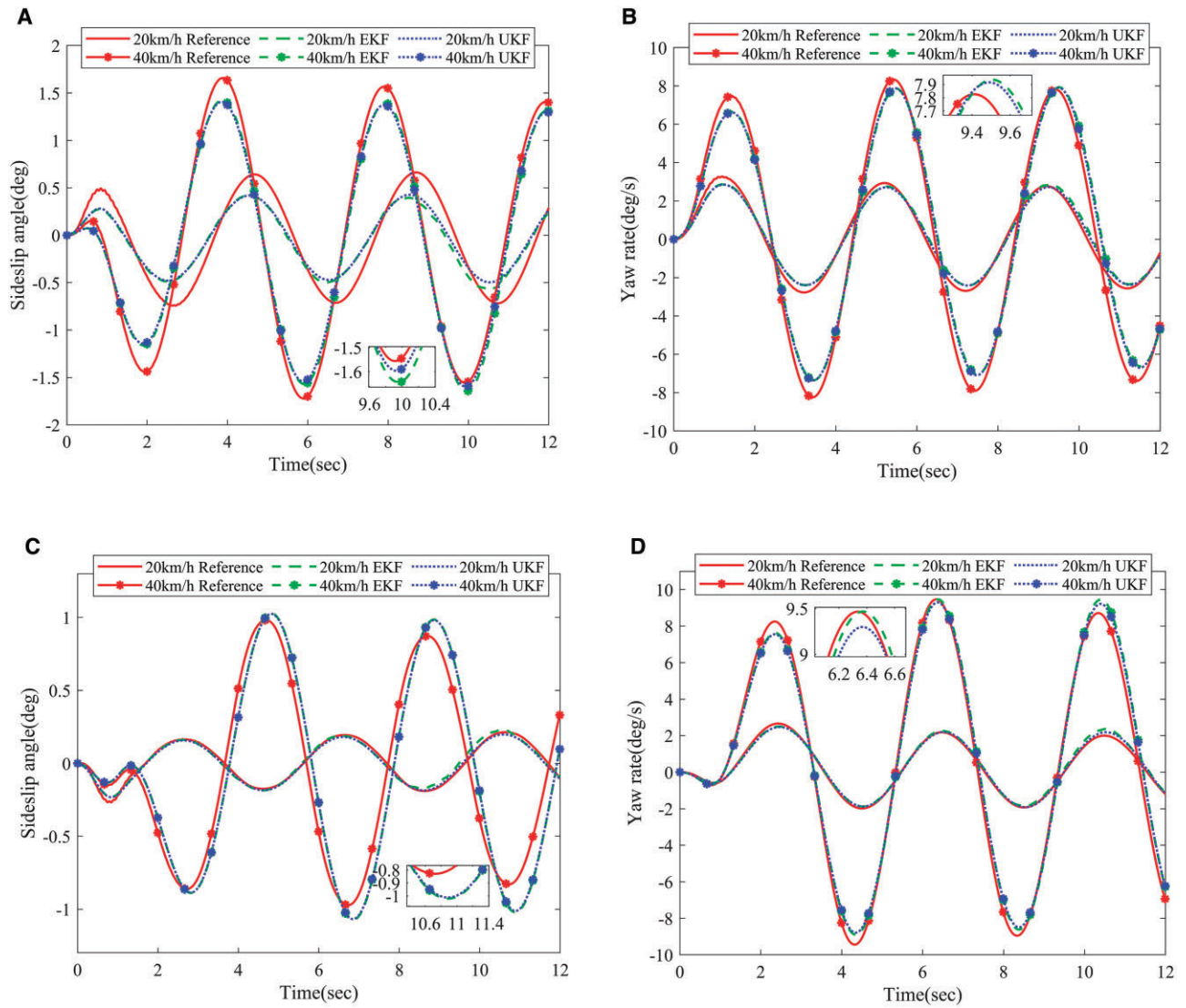


Figure 10: Estimation results based on a 4DOF model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

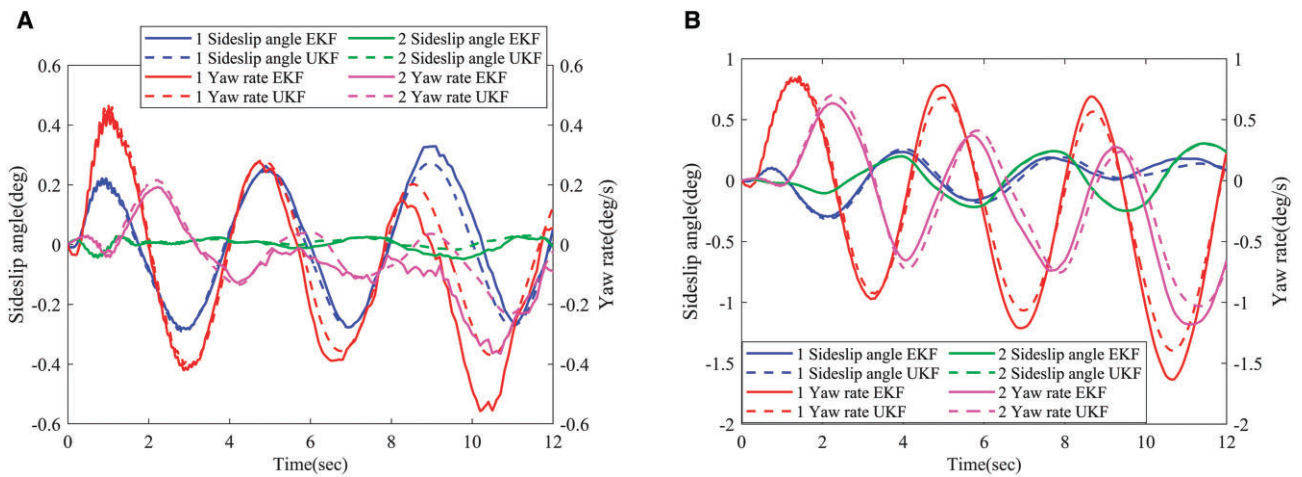


Figure 11: Error plots. (A) The estimation error at 20km/h. (B) the estimation error at 40km/h.

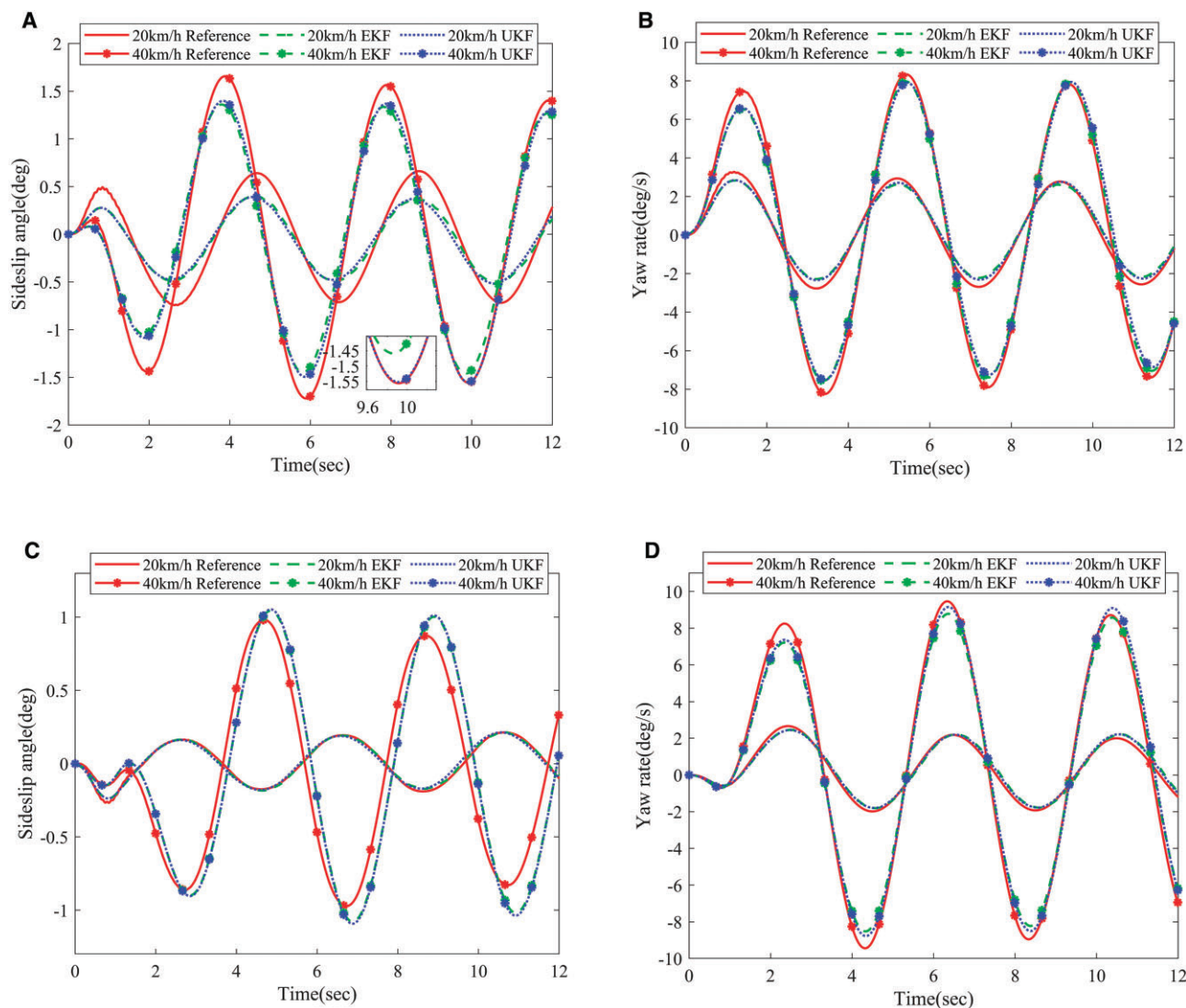


Figure 12: Estimation results based on a 6DOF model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

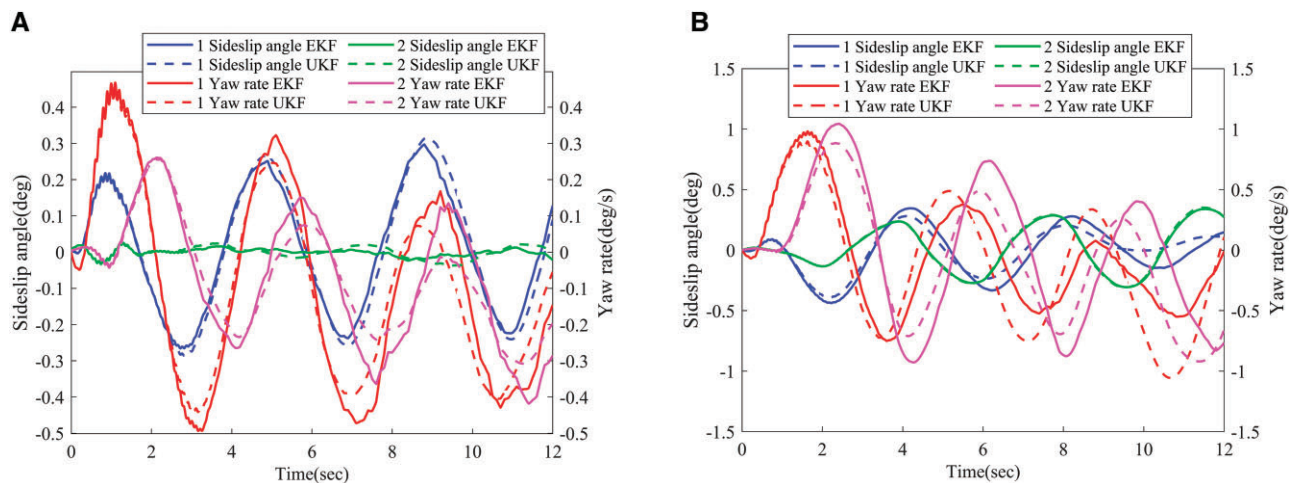


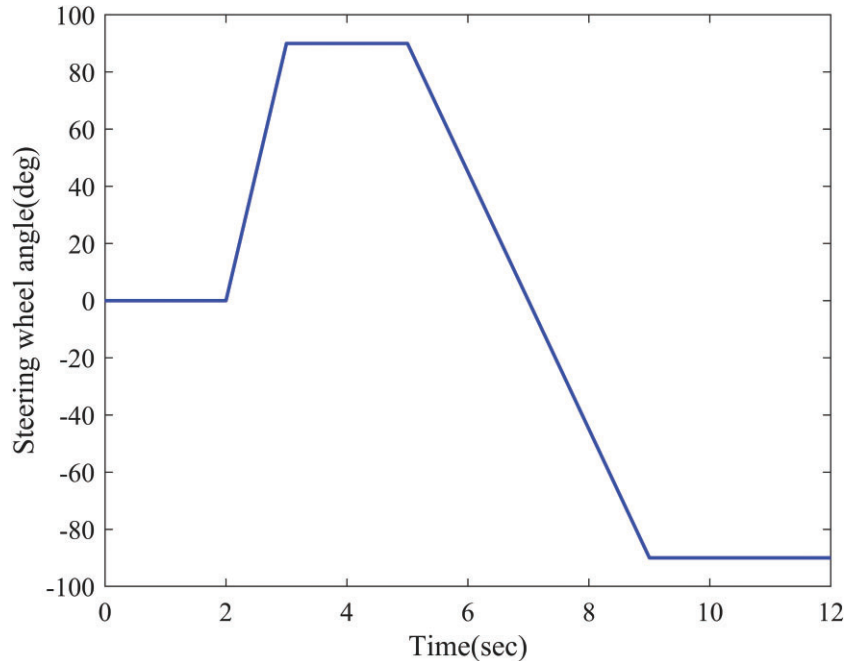
Figure 13: Error plots. (A) The estimation error at 20 km/h. (B) The estimation error at 40 km/h.

Table 2: Error index at 20 km/h under sinusoidal condition.

		MAE				RMSE			
		Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2	Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2
4DOF	EKF	0.1627	0.2318	0.0164	0.1097	0.1860	0.2753	0.0202	0.1420
	UKF	0.1592	0.2112	0.0134	0.0812	0.1809	0.2442	0.0159	0.1041
6DOF	EKF	0.1452	0.2534	0.0075	0.1587	0.1666	0.2919	0.0101	0.1951
	UKF	0.1590	0.2231	0.0137	0.1311	0.1818	0.2614	0.0166	0.1606

Table 3: Error index at 40 km/h under sinusoidal condition.

		MAE				RMSE			
		Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2	Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2
4DOF	EKF	0.1233	0.6501	0.1341	0.4112	0.1439	0.7696	0.1601	0.5249
	UKF	0.1245	0.5846	0.1328	0.4020	0.1477	0.6867	0.1586	0.5017
6DOF	EKF	0.1736	0.4608	0.1597	0.4673	0.2105	0.5442	0.1896	0.5620
	UKF	0.1340	0.3669	0.1628	0.4067	0.1690	0.4465	0.1931	0.4991

**Figure 14:** Steering wheel angle.

resulting in insignificant difference in estimation accuracy with the 6DOF model. In addition, when the speed is low, the system nonlinearity is not very strong. The 4DOF model combined with EKF can improve the calculation efficiency while ensuring accuracy. When the speed is high, the system is strongly nonlinear, and the 6DOF model combined with UKF can achieve the best estimation effect.

5. Experimental Results

In order to verify the actual performance of the estimation strategy studied, this study conducted a semi-physical simulation based on the Logitech G27 driving simulator. The hardware sys-

tem is the Logitech G27 driving simulator, including the steering wheel, pedals and force feedback system, which is connected to the host through a USB interface. In the software system, Trucksim provides a high-precision vehicle dynamics model, and MATLAB/Simulink runs the EKF and UKF state estimation algorithms. The two interact with each other in real time through S-Function. During the experiment, the steering wheel angle signal generated by Logitech G27 is read in real time through the "Joystick Input" module of MATLAB/Simulink, and input into the Trucksim dynamics model to drive the dynamic response of the front wheel steering angle. Trucksim outputs the real state of the vehicle and compares it with the state estimation value in MATLAB/Simulink. The estimation accuracy is quantified by calculating MAE and RMSE. In order to ensure the real-time performance of the sim-

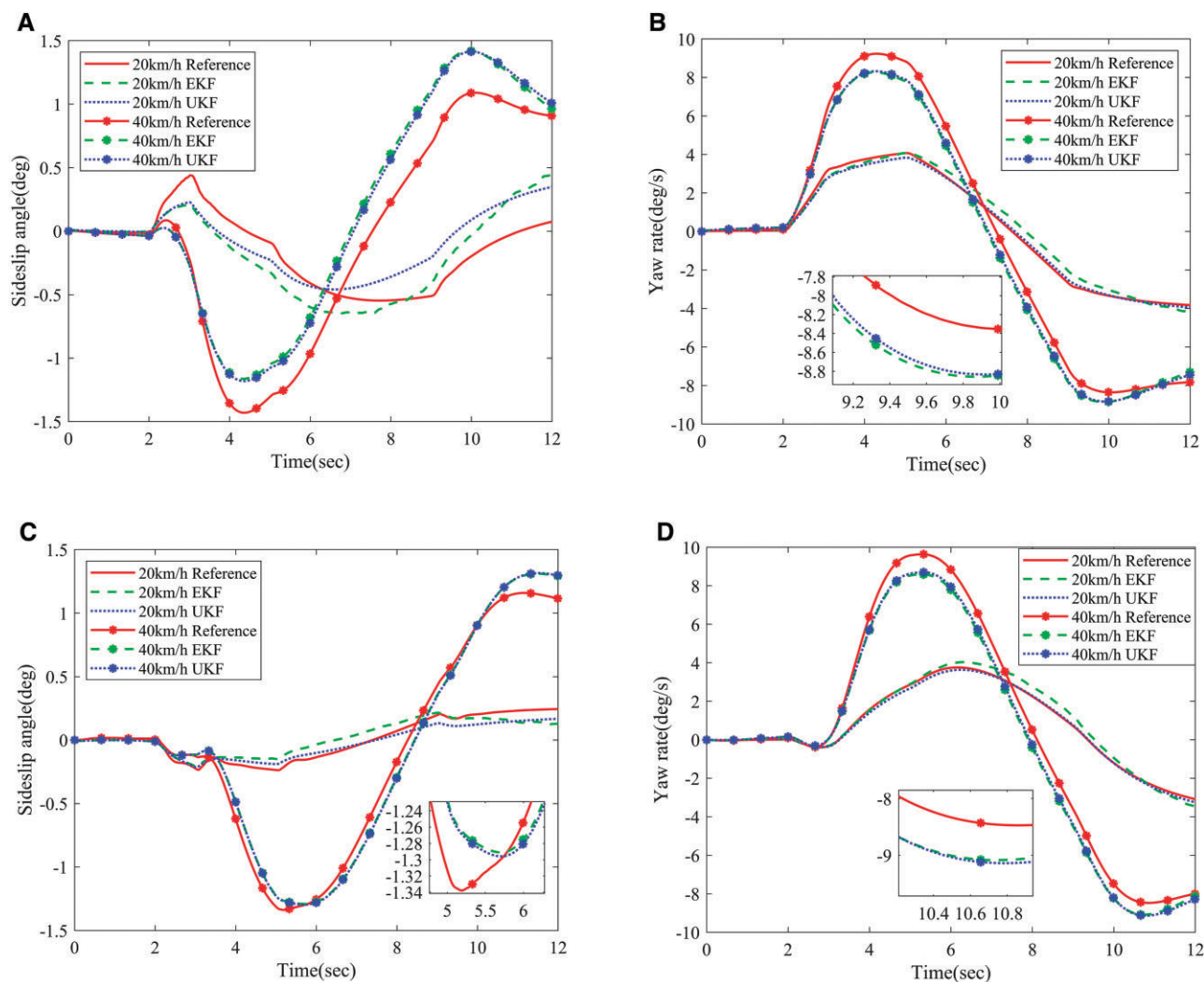


Figure 15: Estimation results based on a 4DOF model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

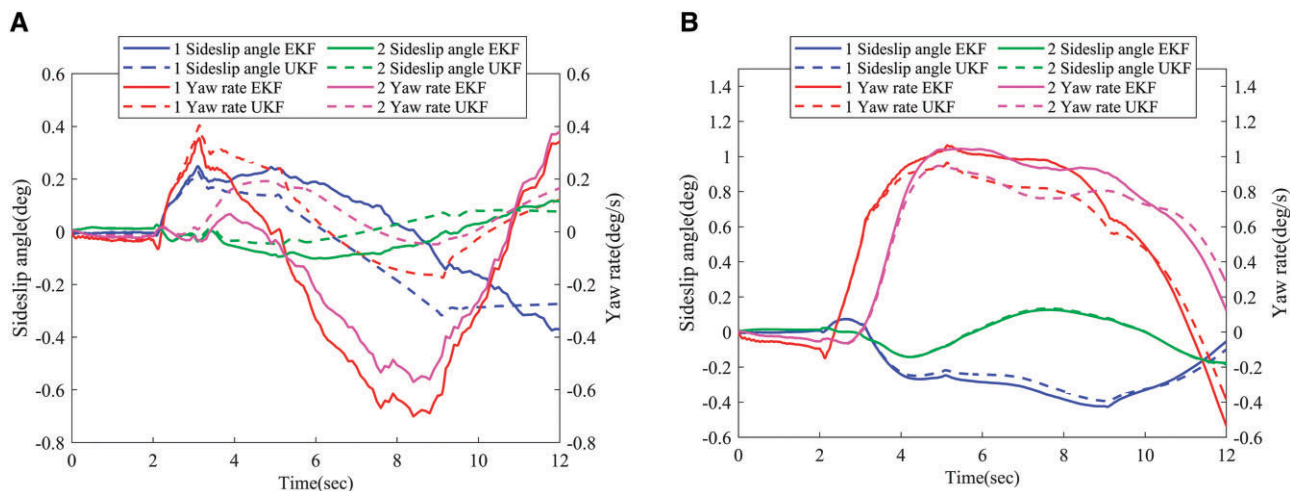


Figure 16: Error plots. (A) The estimation error at 20 km/h. (B) The estimation error at 40 km/h.

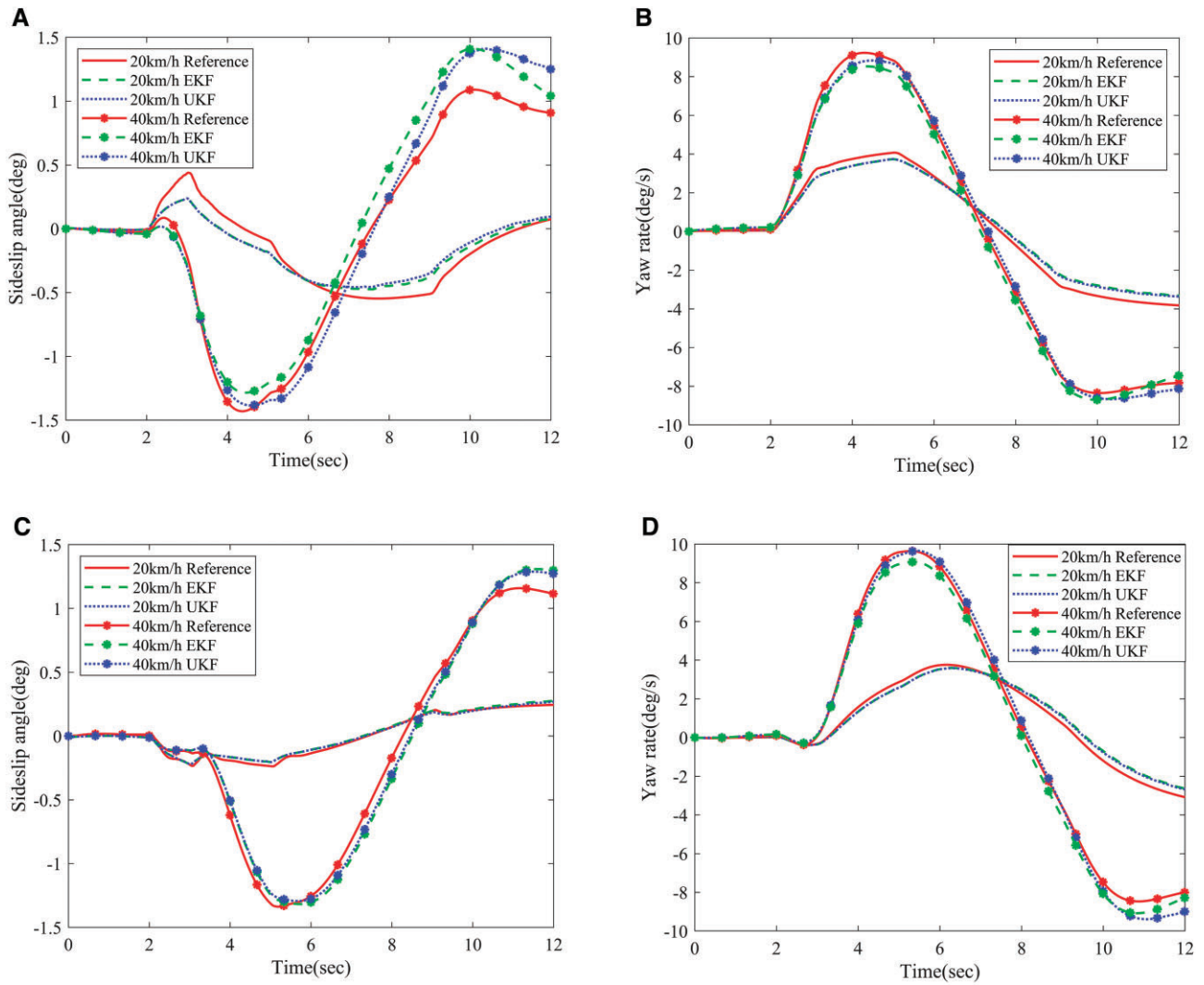


Figure 17: Estimation results based on a 6DOF model. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

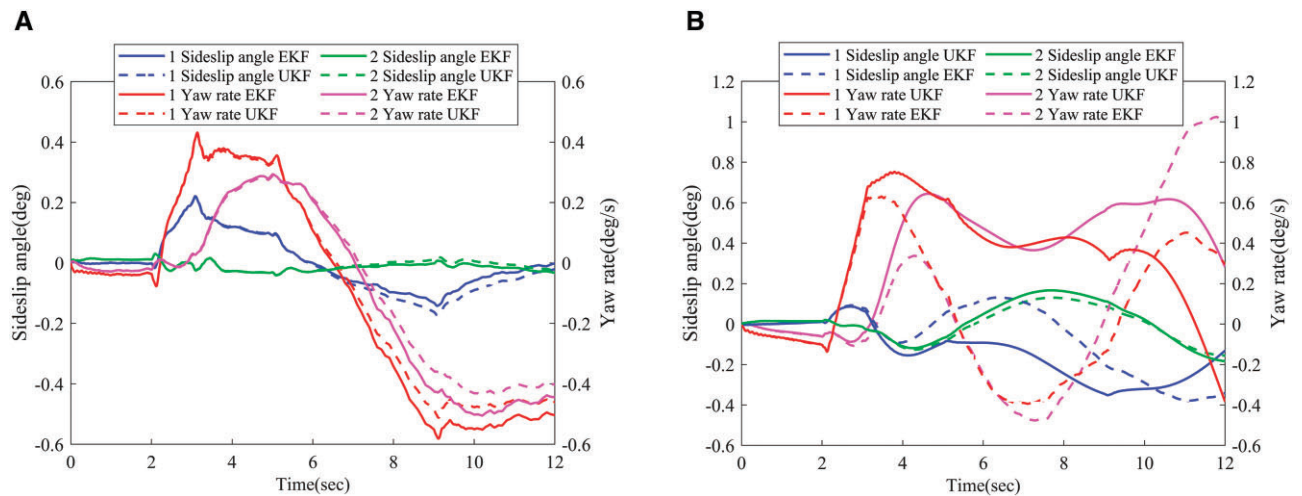


Figure 18: Error plots. (A) The estimation error at 20 km/h, (B) the estimation error at 40 km/h.

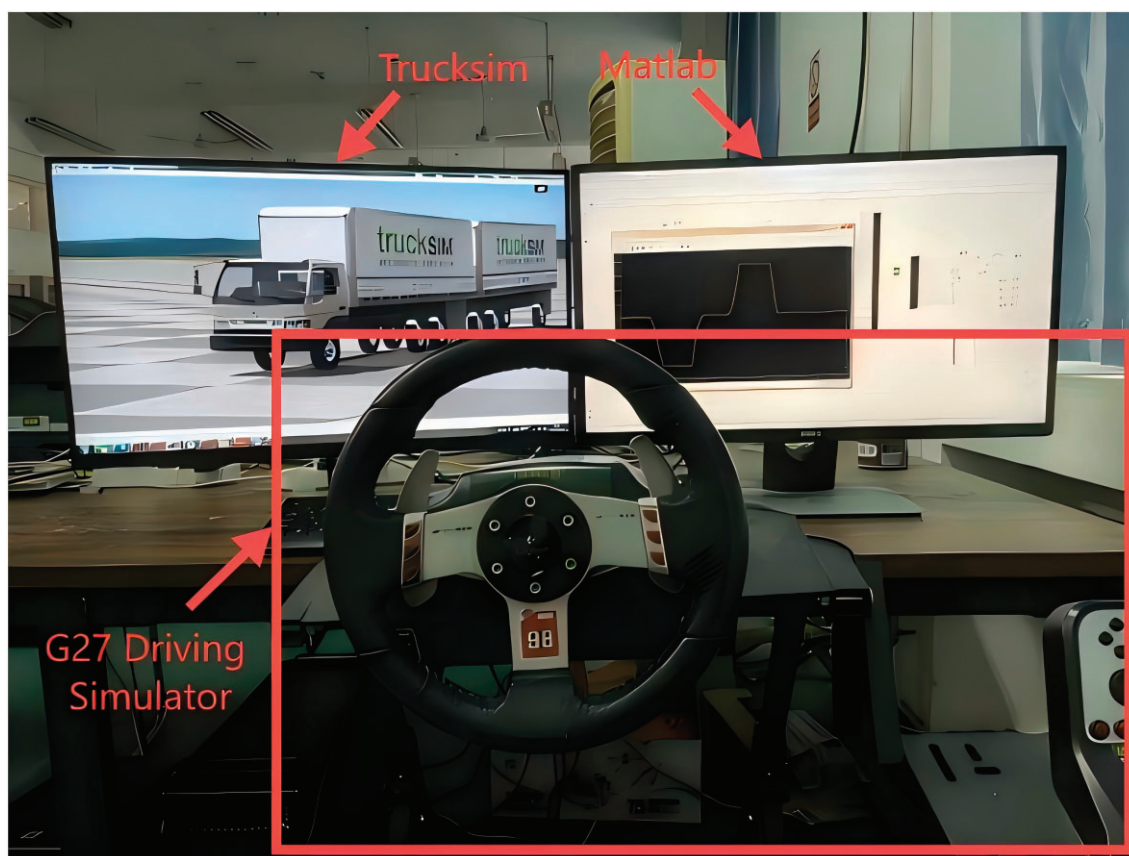


Figure 19: Semi-physical simulation platform.

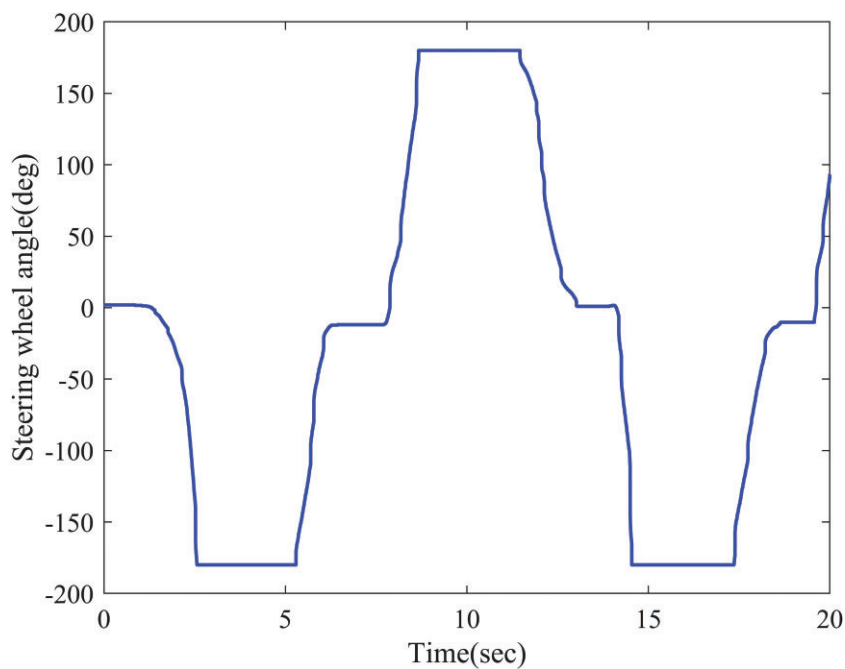


Figure 20: Steering wheel angle.

ulation system, the “Real-Time Sync” module is added to achieve high-precision time synchronization. The hardware timer is used to trigger the synchronous operation of Trucksim and Simulink to ensure that the two are carried out under the same time base, effectively suppressing the timing deviation caused by calcula-

tion delay or communication jitter. The verification analysis is performed using the semi-physical simulation platform shown in Figure 19.

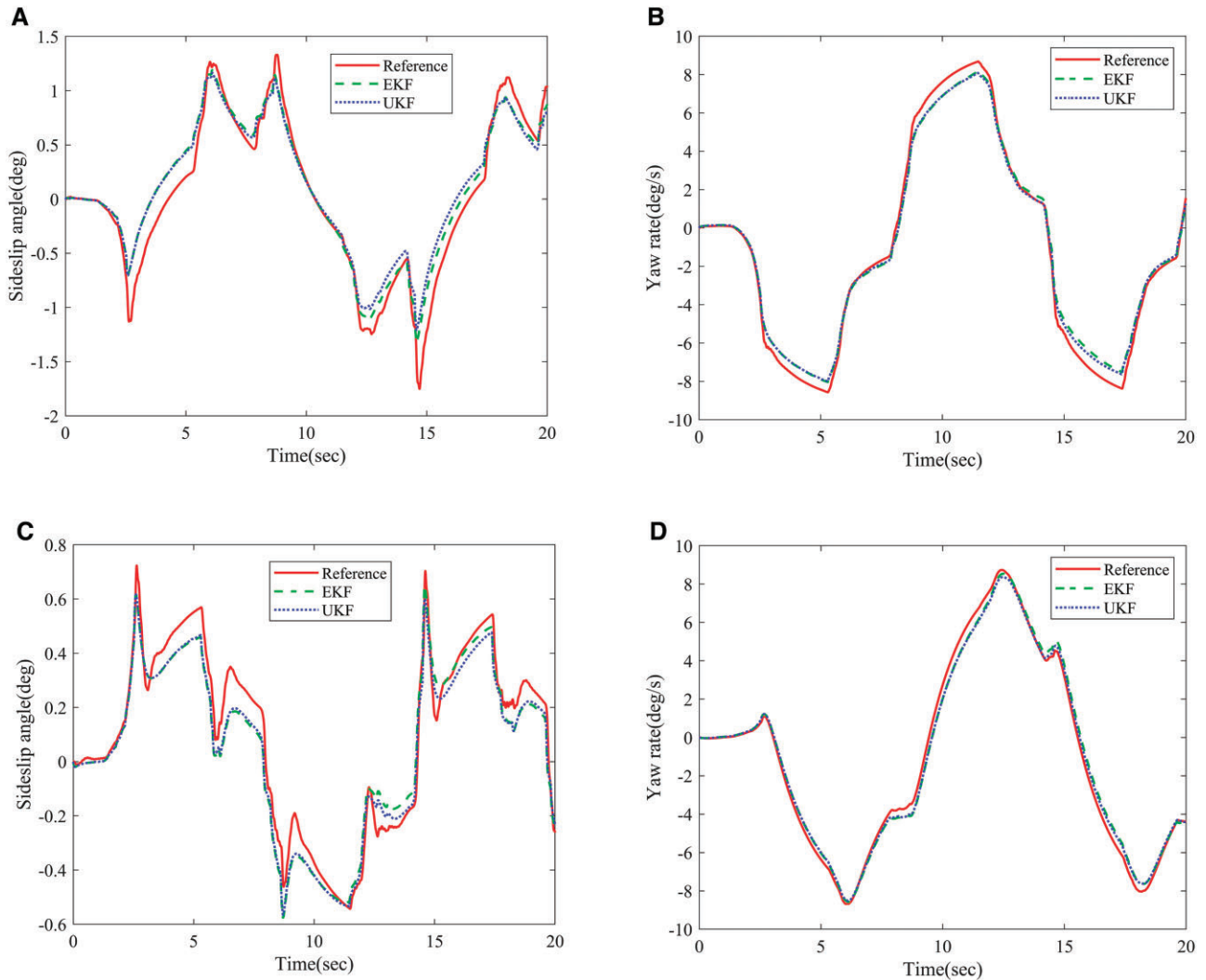
According to the simulation analysis results in the previous section, the state estimator based on the 6DOF model has the

Table 4: Error index at 20 km/h under fishhook condition.

		MAE				RMSE			
		Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2	Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2
4DOF	EKF	0.1437	0.2680	0.0519	0.1918	0.1757	0.3460	0.0625	0.2686
	UKF	0.1478	0.1208	0.0374	0.0690	0.1819	0.1563	0.0457	0.0943
6DOF	EKF	0.0644	0.2939	0.0158	0.2231	0.0836	0.3516	0.0183	0.2846
	UKF	0.0759	0.2689	0.0140	0.2007	0.0943	0.3185	0.0170	0.2528

Table 5: Error index at 40 km/h under fishhook condition.

		MAE				RMSE			
		Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2	Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2
4DOF	EKF	0.2140	0.5858	0.0686	0.5933	0.2566	0.7005	0.0858	0.7160
	UKF	0.1999	0.5236	0.0706	0.5438	0.2372	0.6218	0.0884	0.6441
6DOF	EKF	0.1510	0.3524	0.0650	0.3698	0.1885	0.4122	0.0807	0.4330
	UKF	0.1250	0.2702	0.0753	0.3030	0.1741	0.3192	0.0946	0.4203

**Figure 21:** Semi-physical simulation results. (A) Unit 1's sideslip angle. (B) Unit 1's yaw rate. (C) Unit 2's sideslip angle. (D) Unit 2's yaw rate.

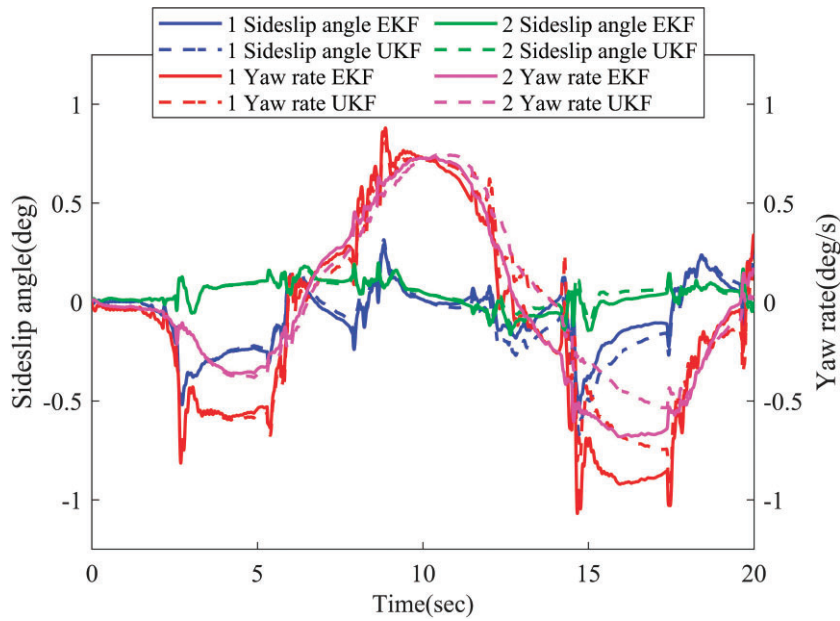


Figure 22: Error plot.

Table 6: Error indicators during semi-physical simulation.

		MAE				RMSE			
		Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2	Sideslip angle of unit 1	Yaw rate of unit 1	Sideslip angle of unit 2	Yaw rate of unit 2
6DOF	EKF	0.1235	0.4246	0.0634	0.3471	0.1671	0.5192	0.0778	0.4221
	UKF	0.1486	0.3886	0.0594	0.3127	0.1960	0.4703	0.0713	0.3822

best estimation effect. Therefore, a semi-physical simulation verification was carried out on the EKF vehicle state estimator and UKF vehicle state estimator based on the 6DOF dynamic model. The speed was selected as 30 km/h and the road adhesion coefficient was 0.85. The steering wheel was turned clockwise from the initial position to 180° , kept for 2–3 seconds and then returned to the center. Then, it was turned counterclockwise to 180° and kept for 2–3 seconds and then returned to the center. This formed a complete “steering-returning” operation. The total simulation time was 20 seconds. The steering wheel angle is shown in Figure 20, and the MAE and root RMSE were calculated.

The semi-physical simulation results are shown in Figure 21, the error plot is shown in Figure 22, and the error indicators are shown in Table 6. According to the simulation findings above, the EKF and UKF vehicle state estimators have a high estimate accuracy in the semi-physical simulation, allowing for precise prediction of the sideslip angle and yaw rate of units 1 and 2. For the estimation of the sideslip angle of the center of mass, both EKF and UKF showed excellent estimation performance, but for the estimation of the yaw rate, the estimation accuracy of the UKF vehicle state estimator was higher than that of the EKF state estimator, with the maximum reduction of the MAE by 9.91% and the maximum reduction of the RMSE by 9.45%.

6. Conclusion

This study aims at the state estimation problem of multi-axis distributed electric drive articulated vehicles. By establishing 4DOF and 6DOF dynamic models and designing state estimators based

on EKF and UKF, the influence of different models and algorithms on the estimation accuracy of center of mass sideslip angle and yaw rate is systematically studied. The research results show that under low-speed weak nonlinear conditions, the accuracy of 4DOF and 6DOF models is similar, while under high-speed strong nonlinear conditions, the 6DOF model significantly improves the estimation accuracy due to the introduction of the roll degree of freedom, with the maximum reduction of MAE by 62.6% and the maximum reduction of RMSE by 62.8%. Algorithm comparison shows that UKF performs outstandingly in the estimation of yaw rate, and is superior to EKF in all aspects, with the maximum reduction of MAE by 64% and the maximum reduction of RMSE by 64.9%. However, UKF has poor estimation accuracy for the sideslip angle of the center of mass. The dynamic characteristics of the sideslip angle of the center of mass are directly affected by the lateral force of the vehicle and the linear model of the tire, and the linear characteristics of the system dominate, while EKF is efficient in calculation and performs better. In general, under low-speed weak nonlinear conditions, the 4DOF model combined with EKF can improve the calculation efficiency while ensuring accuracy, while under high-speed strong nonlinear conditions, the 6DOF model combined with UKF can achieve the best estimation effect. Semi-physical simulation further verifies the practical feasibility of the combination of the 6DOF model and the UKF algorithm. The UKF estimation error is reduced by 9.91% at most compared with the EKF. The research results provide a high-precision state estimation method for the intelligent control of multi-axis articulated vehicles, which can reduce the dependence on high-cost sensors. In the future, we can further explore multi-model fusion

strategies under complex working conditions and combine data-driven methods to improve the adaptability of dynamic parameters, while optimizing real-time performance to meet the needs of vehicle embedded systems.

Conflicts of Interest

The authors declare no conflict of interest.

Author Contributions

Boqiang Zhang: Conceptualization, Methodology, Software, Writing—original draft. **Fengcheng Xue:** Writing—original draft. **You Zhou:** Writing—review & editing, Investigation. **Hualiang Tian:** Supervision, Investigation. **Xun Zhang:** Writing—review & editing, Validation.

Funding

This work was supported by the Henan Province key research and development project (Grant Number 231111241100), Henan Province science and technology development plan (Grant Number 242102240103), Talent Project of Henan University of Technology (Grant Number 2023BS079), International science and technology cooperation cultivation project of Henan Province (Grant Number 242102520008), Projects of higher education institutions in Henan Province (Grant Number 22A460010), Cultivation Project of Tuoxin Team in Henan University of Technology (Grant Number 2024TXTD16), Youth Fund of Natural Science Foundation of Henan Province (Grant Number 252300421540).

Data Availability

The Any relevant source code and study data can be available from the corresponding author by appropriate requests.

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Appendix A: Matrix Expression of State Equation in 6DOF

$$\dot{\mathbf{M}} = \begin{bmatrix} M_{11} & M_{12} & 0 & M_{14} & 0 & 0 & 0 & 0 \\ M_{21} & 0 & 0 & M_{24} & M_{25} & 0 & 0 & M_{28} \\ 0 & 0 & 0 & 0 & M_{35} & M_{36} & 0 & M_{38} \\ 1 & -\frac{d_1}{u_1} & 0 & -\frac{h_{1c}}{u_1} & -1 & -\frac{d_2}{u_2} & 0 & \frac{h_{2c}}{u_2} \\ M_{51} & M_{52} & 0 & M_{54} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & M_{65} & M_{66} & 0 & M_{68} \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad (A1)$$

where $M_{11} = dm_1u_1$, $M_{12} = I_{21}$, $M_{14} = -dm_{s1}h_1 - I_{1xz}$, $M_{21} = m_1u_1$, $M_{24} = -m_{s1}h_1$, $M_{25} = m_2u_2$, $M_{28} = -m_{s2}h_2$, $M_{35} = dm_2u_2$, $M_{36} = -I_{22}$, $M_{38} = I_{2xz} - dm_{s2}h_2$, $M_{51} = h_{1c}m_1u_1 + m_{s1}h_1u_1$, $M_{52} = I_{1xz}$, $M_{54} = -h_{1c}m_{s1}h_1 - I_{1xx} - 2m_{s1}h_1^2$, $M_{65} = h_{2c}m_2u_2 + m_{s2}h_2u_2$, $M_{66} = I_{2xz}$, $M_{68} = -h_{2c}m_{s2}h_2 - I_{2xx} - 2m_{s2}h_2^2$.

$$\tilde{\mathbf{P}} = \begin{bmatrix} P_{11} & P_{12} & P_{13} & 0 & 0 & 0 \\ -k_1 & -k_2 & -k_3 & -k_4 & -k_5 & -k_6 \\ 0 & 0 & 0 & P_{34} & P_{35} & P_{36} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ P_{51} & -h_{1c}k_2 & -h_{1c}k_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & -h_{2c}k_4 & -h_{2c}k_5 & -h_{2c}k_6 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (A2)$$

where $P_{11} = -(a+d)k_1$, $P_{12} = (b-d)k_2$, $P_{13} = (c-d)k_3$, $P_{34} = (c-d)k_4$, $P_{35} = (b-d)k_5$, $P_{36} = (-d-a)k_6$, $P_{51} = -h_{1c}k_1$.

$$\tilde{\mathbf{N}} = \begin{bmatrix} N_{11} & N_{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ N_{21} & N_{22} & 0 & 0 & N_{25} & N_{26} & 0 & 0 \\ 0 & 0 & 0 & 0 & N_{35} & N_{36} & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 & 0 \\ N_{51} & N_{52} & N_{53} & N_{54} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & N_{65} & N_{66} & N_{67} & N_{68} \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (A3)$$

where $N_{11} = (a+d)k_1 + (d-b)k_2 + (d-c)k_3$, $N_{12} = \frac{(ad+a^2)k_1 + (b^2-bd)k_2 + (c^2-cd)k_3}{u_1} - dm_1u_1$, $N_{21} = k_1 + k_2 + k_3$, $N_{22} = \frac{ak_1 - bk_2 - ck_3}{u_1} - m_1u_1$, $N_{25} = k_4 + k_5 + k_6$, $N_{26} = \frac{ck_4 + bk_5 - ak_6}{u_2} - m_2u_2$, $N_{35} = (d-c)k_4 + (d-b)k_5 + (a+d)k_6$, $N_{36} = \frac{(cd-c^2)k_4 + (bd-b^2)k_5 - (a^2+ad)k_6}{u_2} - dm_2u_2$, $N_{51} = (k_1 + k_2 + k_3)h_{1c}$, $N_{52} = \frac{(ak_1 - bk_2 - ck_3)h_{1c}}{u_1} - m_{s1}h_1u_1 - m_1u_1h_{1c}$, $N_{53} = K_1 - m_{s1}gh_1$, $N_{54} = C_1$, $N_{65} = (k_4 + k_5 + k_6)h_{2c}$, $N_{66} = \frac{(ck_4 + bk_5 - ak_6)h_{2c}}{u_2} - m_{s2}h_2u_2 - m_2u_2h_{2c}$, $N_{67} = K_2 - m_{s2}gh_2$, $N_{68} = C_2$.